Tariff Policy with an Exorbitant Privilege*

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Abstract

We study tariffs in a reserve-currency economy where foreign demand for the dominant transaction asset finances a persistent trade deficit for the issuer and shrinks its tradables sector over time. In this environment, a unilateral import tariff cannot reduce the net real transfer that reserve-currency status finances: the long-run quantity of home imports driven by foreign asset demand is invariant to the tariff. In the benchmark, this also leaves tradables employment and output unchanged; with imperfect substitutability between home and foreign goods, tariffs further reduce home tradables employment. The tariff's main effects are on relative prices, fiscal revenue, and the measured trade-deficit-to-GDP ratio, which falls through a valuation effect rather than real trade rebalancing. This neutrality extends to a multi-country setting with a competing settlement currency and endogenous portfolio choice.

JEL Classification: F13, F32, F41, E42.
Keywords: reserve currency; exorbitant privilege; tariffs; trade deficit.

1 Introduction

The United States runs a persistent trade deficit, and tariffs are frequently proposed as the remedy. Whether tariffs can reduce that deficit, however, depends on what the deficit represents. We argue that, in a reserve-currency economy, a persistent trade deficit is the equilibrium counterpart of foreign demand for domestic transaction assets. If that is the relevant margin, then the question is not whether tariffs can alter import prices, but whether they can overturn the foreign demand for the dominant currency. In our framework, the answer is no.

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We study this in what we call an "exorbitant privilege" environment. A country whose liabilities are valued as safe and liquid stores of value can finance a real resource transfer from the rest of the world. Foreigners accumulate the dominant transaction asset by shipping tradable goods in exchange.[1] In the benchmark (Sections 2–3), the exorbitant privilege operates specifically as a seigniorage transfer on fiat balances: foreigners hold the home currency as their only store of value, and the issuer extracts real resources through money growth. Section 5 relaxes this by introducing a competing foreign settlement instrument and endogenous portfolio choice, showing that the neutrality result extends to this richer asset-demand environment. That transfer has real allocation consequences: it changes relative prices between tradables and nontradables and, through those prices, shifts labor away from the home tradables sector. To the extent that tradables are concentrated in industrial goods and services, reserve-currency status can generate a form of deindustrialization.

This perspective differs from a prominent view in current policy debates, where the trade deficit is treated as a price distortion and tariffs as the natural corrective instrument. Recent proposals associated with Miran (2024) and the so-called "Mar-a-Lago Accord" are examples of this broader argument. In that view, broad import tariffs redirect spending toward domestic producers, shrink the import bill, and perhaps force some of the burden onto foreign exporters. That logic is intuitive in partial equilibrium. The question we ask is whether it survives in a general equilibrium in which the trade deficit is pinned down by foreign asset demand rather than by a mispriced import wedge.

We study that question in a stylized two-country dynamic general equilibrium model. The two countries are otherwise identical, except that the home country issues the only internationally accepted transaction asset. Households consume a tradable good and a nontradable service, and production is Ricardo–Viner: each sector has a sector-specific factor and a mobile factor that can move across sectors. The key distinction is between the goods-price margin, which tariffs affect, and the asset-demand margin, which determines the real transfer associated with reserve-currency status. We use a labor-supply/labor-demand representation to make the sectoral implications of that distinction transparent.

The central result is that a unilateral import tariff cannot reduce the net real transfer associated with reserve-currency status. On the foreign side, the quantity of tradables exported to the home country is governed by demand for the home transaction asset, not by the tariff wedge.

[1] In reality, reserve-currency accumulation operates along two distinct margins: a trade-based channel, associated primarily with official reserve management, and a portfolio-based channel, driven by private demand for safe assets. Our benchmark model isolates the former mechanism. Allowing for the latter would weaken the tight link between reserve demand and net exports, thereby attenuating the magnitude of the implied trade imbalances and sectoral reallocation. The multi-country extension in Section 5.3 introduces a competing settlement currency and endogenous portfolio choice, partially addressing this limitation.

This pins down the long-run quantity of home imports independently of the tariff. In the benchmark, the invariance of this transfer also leaves tradables employment and output unchanged; in extensions with differentiated goods, gross trade flows may shift, but the net transfer remains invariant.

On the home side, the tariff raises the consumer price of imported goods, but if tariff revenue is rebated lump-sum to home consumers, their purchasing power rises one-for-one. The relative return to producing tradables at home is therefore unchanged, so long-run tradables employment and output are unchanged as well.

What the tariff does change is valuation. A higher tariff drives a wedge between the price paid by home consumers and the price received by foreign producers. Because trade statistics value imports at world prices while domestic GDP is measured at domestic prices, this wedge mechanically lowers the measured trade-deficit-to-GDP ratio even when import quantities do not change. In this sense, the tariff is a valuation device rather than a real rebalancing device.

The tariff also redistributes nominal income. Foreign exporters receive lower net-of-tariff prices, home consumers pay more for imported goods, and the home government collects the difference. In the benchmark, because tariff revenue is rebated and real allocations are unchanged, this redistribution is purely nominal—no agent's real consumption or utility is affected. Tariffs are not irrelevant in this environment: they raise revenue and alter nominal prices. But they do not eliminate the real trade deficit or reindustrialize the home economy, because those outcomes are governed by asset demand rather than by the border price of imports.

We also show that this neutrality result is not an artifact of the benchmark assumption that the home transaction asset is the only store of value. In a multi-country extension with differentiated goods, a competing foreign settlement instrument, and endogenous portfolio choice between currencies, the same logic survives. A goods tariff operates on the goods-price margin, while the trade deficit is pinned down by the asset-return margin that governs portfolio allocation. By contrast, a financial tariff on foreign reserve holdings operates directly on that asset-return margin and therefore does reduce the trade deficit and expand home tradables employment. The broader message is that, in a reserve-currency economy, the effectiveness of policy depends on which margin it targets.

1.1 Related literature

Our paper is related to three strands of literature.

The first studies the macroeconomic consequences of the dollar's international role. Gourinchas and Rey (2005, 2013) document that the United States functions as a world banker and that external adjustment occurs through valuation effects as well as trade flows. Caballero et al. (2017,

2016) rationalize persistent imbalances as an equilibrium response to global safe-asset scarcity, while Krishnamurthy and Vissing-Jorgensen (2012) document the convenience yield on U.S. Treasuries. The closest theoretical predecessor is Farhi and Maggiori (2018), who study a reserve-currency issuer in the international monetary system. We share their core asymmetry, but focus on a two-sector environment in which reserve-currency demand translates into sectoral labor reallocation and long-run deindustrialization. Bernanke's (2005) global saving glut hypothesis and Obstfeld and Rogoff (2005) provide related perspectives on persistent external imbalances. Gopinath et al. (2020) highlight a complementary channel through dominant-currency pricing.

The second strand studies tariffs, terms of trade, and trade balances. The Lerner symmetry theorem (Lerner, 1936) limits what broad-based tariffs can achieve without accompanying shifts in saving or investment, and the optimal-tariff literature (Metzler, 1949; Bagwell and Staiger, 1999, 2002) emphasizes terms-of-trade effects for large countries. Our model shows how those classic insights interact with reserve-currency asymmetry. Terms-of-trade movements remain important, but in our environment they primarily operate through valuation and revenue redistribution rather than through long-run quantity rebalancing. The Ricardo–Viner production structure (Jones, 1971) makes the sectoral consequences of that interaction transparent.

The third strand emphasizes valuation, incidence, and sectoral adjustment. Caliendo et al. (2025) and Itskhoki and Mukhin (2025) stress valuation channels in interpreting tariff-induced movements in measured trade deficits. Amiti et al. (2019) and Fajgelbaum et al. (2020) document substantial pass-through of tariffs into domestic prices and shared incidence between domestic consumers and foreign exporters. Autor et al. (2013) show that import competition induces persistent local labor-market adjustment away from tradables. Our contribution is to connect these ideas in a reserve-currency setting: the real quantity of net imports is pinned down by asset demand, tariffs move measured deficits through valuation, and the long-run contraction of home tradables employment reflects reserve-currency demand rather than a correctable import-price distortion.

In complementary and independent work, Bigio (2026) studies tariff policy in a reserve-currency economy with capital mobility and money-in-the-utility. Although his mechanism differs from ours, his analysis also points to the limited usefulness of goods tariffs relative to financial policy. Our paper differs by emphasizing sectoral reallocation, deindustrialization, and the distinction between goods-price and asset-return margins in an OLG environment with two production sectors.

Road map. Section 2 develops the model and characterizes the reserve-currency equilibrium. Section 3 studies tariff policy in the benchmark environment, including the neutrality result, the valuation channel, and tariff incidence. Section 4 contrasts tariffs with a production subsidy to

the home tradables sector. Section 5 presents extensions on imperfect substitution, alternative fiscal policy, and a multi-country setting with endogenous portfolio choice. Section 6 concludes. Proofs and additional extensions are collected in the appendix.

2 Model

2.1 Closed Economy

We begin with a closed economy. Agents live for two periods. The young work; the old consume. There are two production sectors—goods and services—and three types of workers. Type 1 is specific to goods, type 2 is specific to services, type 3 is mobile and can work in either. There is one unit of each type.

Let c_t denote services and let k_t denote goods. Let $0 \leq n_t \leq 1$ denote the measure of type 3 (mobile) labor employed in the goods sector. Goods are produced using input from the type 1 worker (fixed) and input from the type 3 worker according to the production function $k_t = n_t^{\alpha}$. Since input of type 1 is fixed, we suppress it as an argument and treat it as a factor in fixed supply. Services are produced using input from type 2 and type 3 workers according to $c_t = G(1 - n_t)$. Let p_t^c and p_t^k denote the money price of services and goods, respectively, and let w_t^3 be the wage of the mobile worker. Indifference of the mobile worker across sectors requires

$$w_t^3 = p_t^c G'(1 - n_t) = p_t^k F'(n_t)$$

For simplicity, assume $G = F$ and $F(n) = n^{\alpha}$, $0 < \alpha < 1$, so that $F'(n) = \alpha F(n)/n$ and $F(n) - nF'(n) = (1 - \alpha) F(n)$.[2] The wages paid to each type are then

$$w_t^1 = p_t^k (1 - \alpha) n_t^{\alpha} \tag{1}$$

$$w_t^2 = p_t^c (1 - \alpha)(1 - n_t)^{\alpha} \tag{2}$$

$$w_t^3 = p_t^k \alpha n_t^{\alpha - 1} = p_t^c \alpha (1 - n_t)^{\alpha - 1} \tag{3}$$

Individuals have identical preferences over future goods and services,

$$U_t = u(c_{t+1}, k_{t+1}).$$

Young households do not value current consumption, so they save all their income. Type 1 and 2 households have a trivial labor supply choice. Type 3 households are indifferent in equilibrium

[2] Different types of labor are not substitutes for one another.

across sectors and earn the same wage w_t^3 in either one.[3]

Goods and services are non-storable. The single asset is nominal government debt (fiat money). Let M_t denote the money supply at date t, where $M_t = \mu M_{t-1}$ with M_0 in the hands of the initial old. New money $\tau_t = (1 - 1/\mu) M_t$ is injected as a lump-sum transfer to the old and divided equally among types. Let m_t^i be young household i's money holdings.

Definition 1. *A closed economy competitive equilibrium is a sequence of consumption allocations and money holdings* $\{c_t^i, k_t^i, m_t^i\}_{t=0,i\in\{1,2,3\}}^{\infty}$, *a fraction of type 3 households in the goods sector* $\{n_t\}_{t=0}^{\infty}$, *prices* $\{p_t^c, p_t^k\}_{t=0}^{\infty}$, *wages* $\{w_t^i\}_{t=0,i\in\{1,2,3\}}^{\infty}$, *and a government policy* $\{M_t, \tau_t\}_{t=0}^{\infty}$, *such that*

1. *Given prices, wages and government policy, allocation* $\{c_t^i, k_t^i\}_{t=0}^{\infty}$ *solves the decision problem of worker* i*:* $\max u\left(c_{t+1}^i, k_{t+1}^i\right)$ *subject to:*

$$m_t^i \leq w_t^i \text{ and } p_{t+1}^c c_{t+1}^i + p_{t+1}^k k_{t+1}^i \leq m_t^i + \frac{\tau_{t+1}}{3}$$

2. *The wages are determined by equations (1), (2) and (3);*

3. *The government budget constraint holds:* $\tau_t = M_t - M_{t-1}$;

4. *Markets clear:* $\sum_{i=1,2,3} c_t^i = (1 - n_t)^{\alpha}$; $\sum_{i=1,2,3} k_t^i = n_t^{\alpha}$; $\sum_{i=1,2,3} m_t^i = M_t$

In any equilibrium, the following accounting identity must hold:

$$p_t^k n_t^{\alpha} + p_t^c (1 - n_t)^{\alpha} = M_t = w_t^1 + w_t^2 + w_t^3,$$

where the left side is nominal GDP and the right side is national income. With Cobb–Douglas preferences $u(c, k) = c^{1-\phi} k^{\phi}$, household optimization implies:

$$p_t^k k_t^i = \phi \left[w_{t-1}^i + \frac{\tau_t}{3}\right] \text{ and } p_t^c c_t^i = (1 - \phi) \left[w_{t-1}^i + \frac{\tau_t}{3}\right]$$

Since old households spend their entire nominal wealth M_t on goods and services:

$$p_t^k n_t^{\alpha} = \phi M_t \text{ and } p_t^c (1 - n_t)^{\alpha} = (1 - \phi) M_t$$

which implies:

$$\frac{p_t^k}{p_t^c} = \frac{\phi}{1 - \phi} \frac{F(1 - n_t)}{n_t^{\alpha}}$$

From (3):

[3]Technically, the young value their time a little bit, to avoid having them supply labor at zero wage rates.

$$\frac{p_t^k}{p_t^c} = \frac{n_t}{1-n_t}\frac{F(1-n_t)}{n_t^\alpha}$$

Combining the previous two equations yields:

$$n_t = \phi$$

so that,

$$\frac{p_t^k}{p_t^c} = \frac{\phi}{1-\phi}\frac{F(1-\phi)}{F(\phi)} \equiv \xi(\phi)$$

2.1.1 A labor supply–labor demand representation

The closed-economy equilibrium allocation of the mobile factor n_t admits a simple graphical characterization that will be useful later.

Labor supply (sectoral choice). The mobile factor (type 3) is indifferent across sectors in equilibrium. The no-arbitrage condition $w_t^3 = p_t^k F'(n_t) = p_t^c F'(1-n_t)$ implies

$$\frac{p_t^k}{p_t^c} = \left(\frac{1-n_t}{n_t}\right)^{\alpha-1}. \tag{4}$$

We interpret (4) as a *labor supply schedule* to the goods (tradables) sector: a higher relative price p_t^k/p_t^c raises the relative return to working in the goods sector and increases n_t.

Labor demand (expenditure shares). Using old-age expenditure shares, $p_t^k n_t^\alpha = \phi M_t$ and $p_t^c F(1-n_t) = (1-\phi)M_t$. Combining these conditions yields

$$\frac{p_t^k}{p_t^c}\frac{n_t^\alpha}{(1-n_t)^\alpha} = \frac{\phi}{1-\phi}. \tag{5}$$

We interpret (5) as a *labor demand schedule* for the goods sector: for a given n_t, the relative price must adjust so that households devote fraction ϕ of nominal spending to goods.

Equilibrium. The intersection of (4) and (5) delivers $n_t = \phi$.

From nominal GDP we can derive prices:

$$p_t^c = \frac{M_t}{\xi(\phi)\,\phi^\alpha + (1-\phi)^\alpha}$$
$$p_t^k = \frac{\xi(\phi)\,M_t}{\xi(\phi)\,\phi^\alpha + (1-\phi)^\alpha}$$

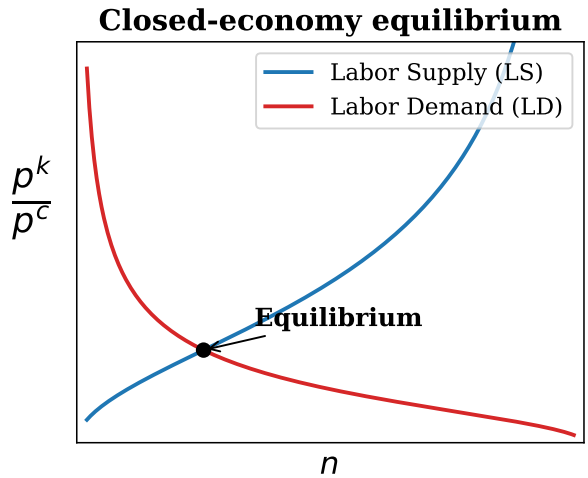


Figure 1: Closed-economy equilibrium as the intersection of a labor supply schedule (4) (blue curve; sectoral choice of the mobile factor) and a labor demand schedule (5) (red curve; expenditure shares). The horizontal axis is the mobile labor share in the goods sector, n, and the vertical axis is the relative price p^k/p^c.

Sector prices grow at rate μ.

Real allocations in this closed economy are invariant to inflation. There is no savings decision and no portfolio allocation problem; inflation scales all nominal wages and prices uniformly. What inflation does is finance the lump-sum transfers to the old—that is its only role here.

2.2 Open Economy

Now add a second country, identical to the first in every respect except one: it has no financial assets. No money, no private financial markets. In autarky this foreign economy cannot function—there is no medium of exchange and hence no production. Open international trade creates an opportunity. The foreign economy has labor and technology; the home economy has the only transaction asset.

What do we expect when trade opens? Goods are tradable; services are not. Home and foreign tradables are perfect substitutes (we relax this in Section 5.1), so the real exchange rate between them is fixed at par. There is a single currency, so there is no nominal exchange rate to worry about (we introduce a second currency with endogenous portfolio choice in Section 5.3).

Let foreign allocations be denoted with an asterisk ($*$). Let p_t^c and p_t^{c*} denote the price of services at home and abroad. Goods are tradable with a common price p_t^k.

Definition 2. *An open economy competitive equilibrium is a sequence of consumption allocations and money holdings in domestic and foreign countries* $\{(c_t^i, k_t^i, m_t^i), (c_t^{i*}, k_t^{i*}, m_t^{i*})\}_{t=0, i\in\{1,2,3\}}^{\infty}$,

the fraction of type 3 households in the goods sector $\{n_t, n_t^*\}_{t=0}^{\infty}$*, prices* $\left\{p_t^c, p_t^{c*}, p_t^k\right\}_{t=0}^{\infty}$*, wages* $\{w_t^i, w_t^{i*}\}_{t=0, i\in\{1,2,3\}}^{\infty}$*, and government policy* $\{M_t, \tau_t\}_{t=0}^{\infty}$*, such that*

1. *Given prices, wages and government policy, allocation* $\{c_t^i, k_t^i, m_t^i\}_{t=0}^{\infty}$ *solves the problem of domestic household* $i = 1, 2, 3$*:* $\max u\left(c_{t+1}^i, k_{t+1}^i\right)$ *subject to:*

$$m_t^i \leq w_t^i \text{ and } p_{t+1}^c c_{t+1}^i + p_{t+1}^k k_{t+1}^i \leq m_t^i + \frac{\tau_{t+1}}{3}$$

2. *Given prices, wages and government policy, allocation* $\{c_t^{i*}, k_t^{i*}, m_t^{i*}\}_{t=0}^{\infty}$ *solves the problem of foreign household* i *(the difference is that they do not receive the transfer):* $\max u\left(c_{t+1}^{i*}, k_{t+1}^{i*}\right)$*, subject to:*

$$m_t^{i*} \leq w_t^{i*} \text{ and } p_{t+1}^{c*} c_{t+1}^{i*} + p_{t+1}^k k_{t+1}^{i*} \leq m_t^{i*}$$

3. *Domestic wages are determined according to equations (1), (2) and (3) (and similarly for foreign wages).*

4. *The government budget constraint holds:* $\tau_t = M_t - M_{t-1}$;

5. *Markets clear*

$$\sum_{i=1,2,3} c_t^i = (1 - n_t)^{\alpha}$$
$$\sum_{i=1,2,3} c_t^{i*} = (1 - n_t^*)^{\alpha}$$
$$\sum_{i=1,2,3} k_t^i + \sum_{i=1,2,3} k_t^{i*} = n_t^{\alpha} + n_t^{*\alpha}$$
$$\sum_{i=1,2,3} m_t^i + \sum_{i=1,2,3} m_t^{i*} = M_t$$

The equilibrium can be reduced to a system of three equations, as the following lemma states.

Lemma 1. *In the open economy, competitive equilibrium is characterized by the following system of equations.*

$$(1 - n_t^*)\, n_t^{*\alpha-1} = (1 - \phi)\, \frac{p_{t-1}^k}{p_t^k} n_{t-1}^{*\alpha-1} \tag{6}$$

$$(1 - n_t)\, n_t^{\alpha-1} = (1 - \phi)\, \frac{p_{t-1}^k}{p_t^k} \left[\mu n_{t-1}^{\alpha-1} + (\mu - 1)\, n_{t-1}^{*\alpha-1}\right] \tag{7}$$

$$n_t^{\alpha} + n_t^{*\alpha} = \phi \frac{p_{t-1}^k}{p_t^k} \mu \left[n_{t-1}^{\alpha-1} + n_{t-1}^{*\alpha-1}\right] \tag{8}$$

and

$$p_0^c\,(1-n_0)^\alpha = (1-\phi)\,M_0 \tag{9}$$

$$p_0^k\,(n_0^{*\alpha} + n_0^\alpha) = \phi M_0 \tag{10}$$

$$\frac{p_0^k}{p_0^c} = \frac{n_0}{1-n_0}\frac{(1-n_0)^\alpha}{n_0^\alpha} \tag{11}$$

with M_0 *given and* $n_0^* = 1$

Proof. See Appendix A.1. □

Interpretation of the equilibrium conditions. Equation (6) is the *foreign service-market clearing condition*: the value of services produced abroad equals the fraction $(1-\phi)$ of foreigners' nominal income, with the ratio p_{t-1}^k/p_t^k capturing goods-price inflation. This equation pins down n_t^* given the history (n_{t-1}^*) and the price path.

Equation (7) is the *home service-market clearing condition.* Home residents spend fraction $(1-\phi)$ of their total nominal income on services. The term μ multiplying home wages reflects money injected each period (seigniorage), while $(\mu-1)$ times foreign wages captures seigniorage collected from foreign money holdings. This extra income is the quantitative heart of the exorbitant privilege.

Equation (8) is *global goods-market clearing*: world tradables production equals world demand, where demand is fraction ϕ of world nominal income (scaled by μ to account for new money).

Together, the three equations determine the sequence $(n_t, n_t^*, p_{t-1}^k/p_t^k)$, tracing out how the global labor allocation and price path co-evolve.

The opening period is especially informative. Home's initial old hold all the money; they can buy goods and services, and production proceeds normally in both sectors. The foreign old have nothing—no money, no claims—and can buy nothing. Since foreign services cannot be traded and no one in the foreign economy can pay for them, no services are produced there in period 0. Every mobile worker in the foreign economy goes into goods production; type 2 workers—specific to services—are idle. The entire foreign goods output is shipped to the home economy in exchange for money. This is the transaction that sets everything in motion, and it is why the initial condition is $n_0^* = 1$. Combining the initial conditions gives

$$\frac{n_0}{1-n_0}\frac{1+n_0^\alpha}{n_0^\alpha} = \frac{\phi}{1-\phi}$$

Importantly, n_0 does not depend on μ. And $n_0 < \phi$: the immediate impact of opening trade is a contraction of home goods production, as the flood of cheap foreign goods lowers the relative

return to producing tradables at home.

Given the initial labor allocation, the initial prices are:

$$p_0^c = \frac{(1-\phi)\, M_0}{(1-n_0)^\alpha}$$
$$p_0^k = \frac{\phi M_0}{1+n_0^\alpha}$$

Let m_t^d and m_t^{d*} be the money that young households hold at the end of period t, with $m_t^d + m_t^{d*} = M_t = \mu^t M_0$. In the initial period:

$$m_0^d = \frac{\phi M_0 n_0^\alpha}{1+n_0^\alpha} + (1-\phi)\, M_0$$

and

$$m_0^{d*} = \frac{\phi M_0}{1+n_0^\alpha}$$

Let GDP_t and GDP_t^* denote nominal GDP in the home and foreign economies, and EXP_t and EXP_t^* aggregate nominal expenditure (total spending by the old, inclusive of seigniorage transfers):

$$\mathrm{GDP}_t = p_t^k n_t^\alpha + p_t^c\,(1-n_t)^\alpha$$
$$\mathrm{GDP}_t^* = p_t^k n_t^{*\alpha} + p_t^{c*}\,(1-n_t^*)^\alpha$$

$$\mathrm{EXP}_t = m_{t-1}^d + (\mu-1)\left(m_{t-1}^d + m_{t-1}^{d*}\right)$$
$$\mathrm{EXP}_t^* = m_{t-1}^{d*}$$

2.2.1 Steady State

Define a steady state as a situation where $n_t = n, n_t^* = n^*$ and $p_t^c/p_{t-1}^c = p_t^{c*}/p_{t-1}^{c*} = p_t^k/p_{t-1}^k = \mu$.

Three properties of the steady state are worth stating before we prove them. First, the reserve-currency asymmetry reallocates mobile labor in opposite directions in the two countries: the foreign economy tilts toward tradables and the home economy toward nontradables. Second, this reallocation changes the relative wage of goods and service producers in opposite directions across countries. Third, the home economy runs a persistent trade deficit. These are not separate phenomena; they are all consequences of the same underlying force: foreign demand for the home transaction asset.

Let us define the wage ratio as the wage of a goods producer relative to that of a service producer.

Proposition 1. *Suppose $\mu \geq 1$. In steady state, the following statements are true:*

(i) $n^* \geq \phi \geq n$, *with equality if and only if* $\mu = 1$;

(ii) $\frac{w_t^1}{w_t^2} \leq \frac{\phi}{1-\phi} \leq \frac{w_t^{1*}}{w_t^{2*}}$, *with equality if and only if* $\mu = 1$;

(iii) $\dfrac{\mathrm{EXP}_t}{\mathrm{GDP}_t} \geq 1 \geq \dfrac{\mathrm{EXP}_t^*}{\mathrm{GDP}_t^*}$, *with equality if and only if* $\mu = 1$.

Proof. See Appendix A.2. □

As long as money grows, the foreign economy produces more tradables than in autarky and ships the excess home in exchange for money. The home economy absorbs those shipments and tilts its mobile labor toward nontradables. The goods-to-services wage ratio falls at home and rises abroad. Trade never balances as long as money is growing: the home economy always consumes more than it produces, paying with claims on future seigniorage. Even if money growth stops, the accumulated foreign money balances remain outstanding—the debt is never paid, it is simply held.

The foreign economy tilts toward tradables because tradable goods are the only object it can deliver in exchange for money. The home economy tilts toward nontradables because access to cheap imported goods lowers the relative profitability—and the relative wage—of producing tradables at home, drawing mobile labor into the nontradables sector. The two effects are the same phenomenon seen from opposite sides.

2.2.2 Steady-state labor supply and labor demand

The steady-state reallocation $n < \phi < n^*$ has a transparent interpretation in terms of a labor-supply/labor-demand diagram, analogous to Figure 1.

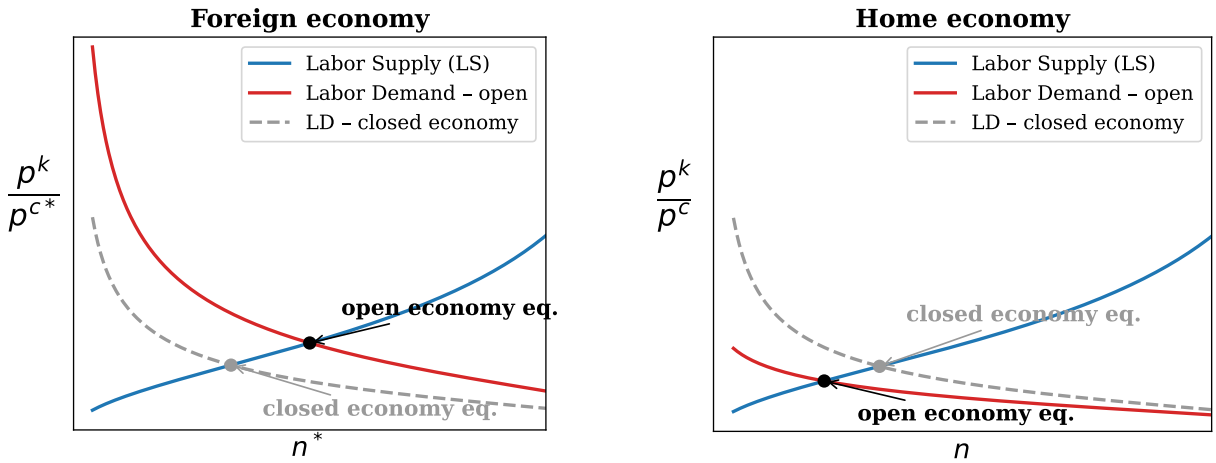
Labor supply (sectoral choice). In both countries, the mobile factor must be indifferent across sectors. This delivers the same upward-sloping "labor supply" schedule as in the closed economy:

$$\frac{p^k}{p^c} = \left(\frac{1-n}{n}\right)^{\alpha-1}, \qquad \frac{p^k}{p^{c*}} = \left(\frac{1-n^*}{n^*}\right)^{\alpha-1}. \tag{12}$$

Labor demand (asset-demand wedge). What differs across countries is the schedule linking relative prices to expenditure shares once the reserve-currency asymmetry is taken into account.

In the foreign economy, steady-state goods-market clearing and the fact that prices grow at rate μ imply the "labor demand" condition

$$\frac{(1-n^*)^\alpha}{n^{*\,\alpha-1}} = \frac{1-\phi}{\mu}\,\frac{p^k}{p^{c*}}. \tag{13}$$



(a) Foreign economy.

(b) Home economy.

Figure 2: Open-economy steady state as an intersection of labor supply (blue curve; (12)) and labor demand (red curve; (13)–(14)). The dashed gray curve is the closed-economy labor demand (5), and the gray dot is the closed-economy equilibrium. The reserve-currency asymmetry shifts the foreign labor-demand schedule up (raising n^* above the closed-economy benchmark) and shifts the home labor-demand schedule down (lowering n below the closed-economy benchmark).

Because the foreign economy must acquire the transaction asset by exporting tradables, the shadow value of producing tradables is high, shifting the foreign labor-demand curve up relative to autarky.

In the home economy, the analogous steady-state condition becomes

$$\frac{p^k}{p^c} = \frac{\phi}{1-\phi} \frac{(1-n)^\alpha}{\left(1-\frac{1}{\mu}\right) n^{*\,\alpha-1} + n^\alpha}. \tag{14}$$

The term $\left(1-\frac{1}{\mu}\right) n^{*\,\alpha-1}$ captures the additional nominal spending power accruing to home residents from money growth on a balance sheet partly held abroad—seigniorage from foreign asset demand. This shifts the home labor-demand curve down, reducing the relative price of tradables and drawing mobile labor into nontradables.

2.2.3 Transition

The steady-state results describe long-run allocations. The transition dynamics, which we cannot characterize analytically, are illustrated in Figure 3 for the parameter values $\phi = 0.2$, $\alpha = 0.5$, $\mu = 1.05$.

The opening period is the most extreme part of the transition. The foreign economy, eager to

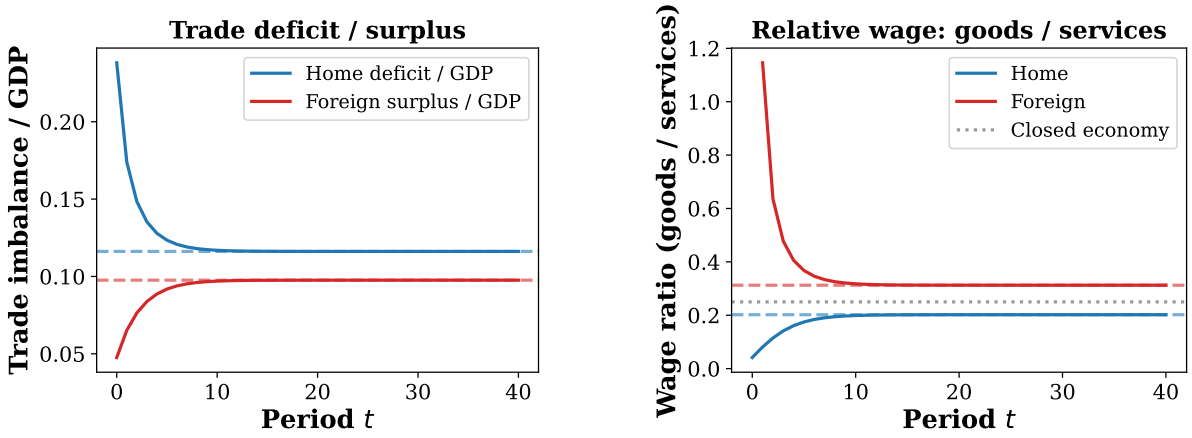


Figure 3: Transition dynamics following the opening of trade ($\phi = 0.2$, $\alpha = 0.5$, $\mu = 1.05$). Left panel: home trade deficit as a share of home goods GDP (blue, declining from an initial peak toward steady state) and foreign trade surplus as a share of foreign goods GDP (red, rising toward steady state). Right panel: wage ratio of goods producers to service producers for the home economy (blue) and foreign economy (red); the dotted gray line marks the common closed-economy benchmark. In the open-economy steady state the home ratio lies below the autarky level (deindustrialization) while the foreign ratio lies above it.

acquire money the moment trade opens, exports its entire goods output to the home economy. The trade deficit jumps immediately to its maximum. If there is no money growth, the foreign economy has all the money it needs after one period and stops exporting; the deficit falls to zero and the economies thereafter operate as two closed economies, the foreign country holding its money balances forever.

With money growth, the foreign economy must continuously export to replenish the real value of money holdings eroded by inflation. The economies transition gradually to the steady state described above. The trade deficit declines from its initial peak but remains positive throughout. For the transition figure, we report the trade imbalance relative to goods GDP. The goods-to-services wage ratio in the home economy follows the same pattern in reverse: it is lowest on impact and rises toward its steady-state value.

3 Tariff Policy

To connect the model to the policy discussion, interpret the home economy as the United States and the foreign economy as the rest of the world. The open-economy equilibrium just described features a structural trade deficit: the dollar is "overvalued" in the sense that it commands more

real goods than it would if the U.S. were not the reserve-currency issuer,[4] and the home tradables sector has contracted relative to the closed-economy benchmark. Both phenomena follow from the same source: foreign demand for U.S.-issued transaction assets. This is what Miran (2024) identifies as the mechanism behind the persistent U.S. trade deficit.

The policy question is whether a unilateral import tariff can undo this. We model the tariff as a proportional tax θ on goods produced abroad and consumed at home. Given perfect substitutability, all trade flows from the foreign economy to the home economy; without loss of generality, foreign consumers consume only foreign-produced goods. Tariff revenue is rebated lump-sum to home old-age consumers.

Let p_t^k and p_t^{k*} be the prices of home- and foreign-produced goods. Let k_{Ht} and k_{Ft} denote home consumption of domestic and foreign goods, and k_{Ft}^* consumption of foreign goods in the foreign economy. In equilibrium $p_t^k = (1+\theta)\, p_t^{k*}$.

Definition 3. *A tariff-distorted competitive equilibrium is a sequence of domestic and foreign allocations and money holdings* $\{(c_t^i, k_{Ht}^i, k_{Ft}^i, m_t^i)\,, (c_t^{i*}, k_{Ft}^{i*}, m_t^{i*})\}_{t=0,\, i\in\{1,2,3\}}^{\infty}$*, mobile-labor allocations* $\{n_t, n_t^*\}_{t=0}^{\infty}$*, prices* $\{p_t^c, p_t^{c*}, p_t^k, p_t^{k*}\}_{t=0}^{\infty}$*, wages* $\{w_t^i, w_t^{i*}\}_{t=0,\, i\in\{1,2,3\}}^{\infty}$*, and government policy* $\{\theta, M_t, \tau_t\}_{t=0}^{\infty}$*, such that*

1. *Given prices, wages and government policy, allocation* $\{c_t^i, k_t^i, m_t^i\}_{t=0}^{\infty}$ *solves the problem of domestic household* i*:* $\max u\left(c_{t+1}^i, k_{Ht+1}^i + k_{Ft+1}^i\right)$*, subject to,*

$$m_t^i \leq w_t^i \text{ and } p_{t+1}^c c_{t+1}^i + p_{t+1}^k k_{Ht+1}^i + (1+\theta)\, p_{t+1}^{k*} k_{Ft+1}^i \leq m_t^i + \frac{\tau_{t+1}}{3};$$

2. *Given prices, wages, and government policy, allocation* $\{c_t^{i*}, k_t^{i*}, m_t^{i*}\}_{t=0}^{\infty}$ *solves the problem of the foreign individual worker* i *(note: the difference is that they do not receive the transfer):* $\max u\left(c_{t+1}^{i*}, k_{Ft+1}^{i*}\right)$ *subject to,*

$$m_t^{i*} \leq w_t^{i*} \text{ and } p_{t+1}^{c*} c_{t+1}^{i*} + p_{t+1}^{k*} k_{Ft+1}^{i*} \leq m_t^{i*}$$

3. *Domestic wages are determined according to equations (1), (2) and (3) (and similarly for foreign wages).*

4. *The government budget constraint holds:*

$$\tau_t = M_t - M_{t-1} + \theta p_t^{k*} \left(\sum_{i=1,2,3} k_{Ft}^i \right)$$

[4] Dollar "overvaluation" here takes the form of a lower price level since there is no nominal exchange rate.

5. *Markets clear:*

$$\sum_{i=1,2,3} c_t^i = (1-n_t)^\alpha$$
$$\sum_{i=1,2,3} c_t^{i*} = (1-n_t^*)^\alpha$$
$$\sum_{i=1,2,3} k_{Ht}^i = n_t^\alpha$$
$$\sum_{i=1,2,3} k_{Ft}^i + \sum_{i=1,2,3} k_{Ft}^{i*} = n_t^{*\alpha}$$
$$\sum_{i=1,2,3} m_t^i + \sum_{i=1,2,3} m_t^{i*} = M_t$$

The equilibrium conditions under the tariff are summarized in the following lemma.

Lemma 2. *Equilibrium allocations and prices in the open economy with a tariff are characterized by the following system of equations for $t > 0$:*

$$(1-n_t^*)\, n_t^{*\alpha-1} = (1-\phi)\, \frac{p_{t-1}^{k*}}{p_t^{k*}} n_{t-1}^{*\alpha-1} \tag{15}$$

$$\begin{aligned}(1-n_t)\, n_t^{\alpha-1} = (1-\phi) \Bigg[& \mu \frac{p_{t-1}^k}{p_t^k} n_{t-1}^{\alpha-1} + (\mu-1)\, \frac{p_{t-1}^{k*}}{p_t^k} n_{t-1}^{*\alpha-1} \\ & + \theta \frac{p_t^{k*}}{p_t^k} \left(n_t^{*\alpha} - \phi \frac{p_{t-1}^{k*}}{p_t^{k*}} n_{t-1}^{*\alpha-1} \right) \Bigg]\end{aligned} \tag{16}$$

$$\begin{aligned} n_t^\alpha + n_t^{*\alpha} = \phi \Bigg[& \mu \frac{p_{t-1}^k}{p_t^k} n_{t-1}^{\alpha-1} + (\mu-1)\, \frac{p_{t-1}^{k*}}{p_t^k} n_{t-1}^{*\alpha-1} \\ & + \theta \frac{p_t^{k*}}{p_t^k} \left(n_t^{*\alpha} - \phi \frac{p_{t-1}^{k*}}{p_t^{k*}} n_{t-1}^{*\alpha-1} \right) \Bigg] + \phi \frac{p_{t-1}^{k*}}{p_t^{k*}} n_{t-1}^{*\alpha-1}\end{aligned} \tag{17}$$

and

$$p_t^k = (1+\theta) p_t^{k*}$$

Proof. See Appendix A.3. □

The key observation is equation (15): it is identical to the no-tariff foreign condition (6). The tariff does not appear in the foreign equilibrium condition at all. This is not a coincidence. The foreign economy's decisions—how much to produce, how much to export, and how much money to hold—are governed entirely by its demand for the home transaction asset. None of these mar-

gins depend on what the home country charges at the border. As a result, θ drops out of the foreign steady state entirely.

In steady state, imposing the steady-state condition in equation (15) gives

$$n^* = \frac{\mu - 1}{\mu} + \frac{\phi}{\mu}.$$

This is independent of the tariff rate. Imposing a tariff on imports does not affect the foreign steady-state allocation.

Turning to the home economy: imposing the steady-state condition and substituting $p_t^k = (1+\theta)p_t^{k*}$ in equation (16) and simplifying (substituting for n^* and rearranging) yields

$$\phi n^{\alpha-1} - n^{\alpha} = (1-\phi)\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1}$$

This too is independent of θ. The home labor allocation is also invariant to the tariff. Long-run sectoral employment and output are unchanged.

Let $K_F \equiv \sum_{i=1,2,3} k_{Ft}^i$ denote aggregate home imports of foreign-produced goods.

Proposition 2 (Long-Run Tariff Neutrality)**.** *Suppose $\mu > 1$. For any tariff rate $\theta \geq 0$, the steady-state allocation (n, n^*, K_F) is invariant to θ. In particular:*

(i) $n^* = 1 - (1-\phi)/\mu$, *independently of θ;*

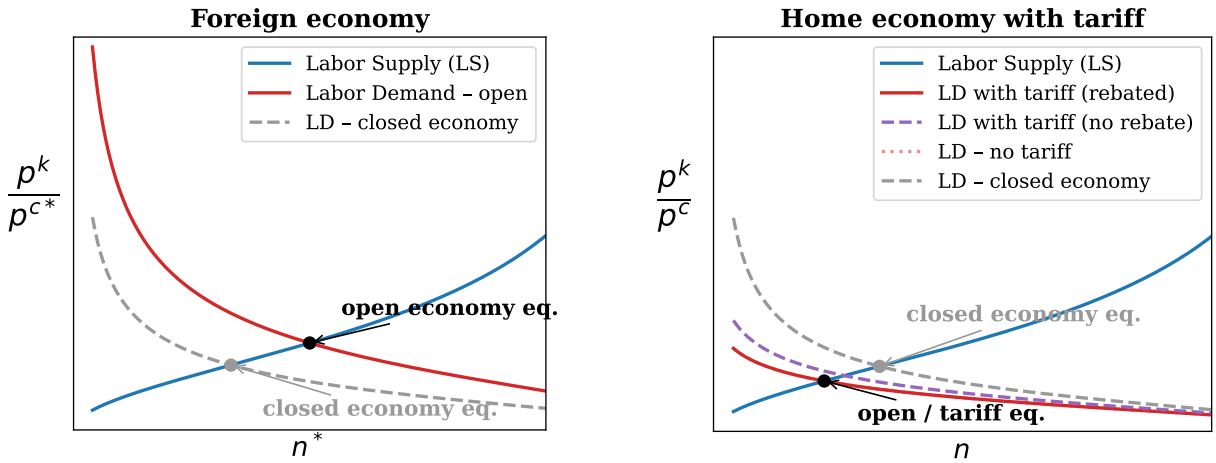
(ii) n *satisfies* $\phi n^{\alpha-1} - n^{\alpha} = (1-\phi)\big[(\mu-1)/\mu\big] n^{*\,\alpha-1}$, *independently of θ;*

(iii) $K_F = \big[(\mu-1)/\mu\big] n^{*\,\alpha-1}$, *independently of θ.*

Proof. See Appendix A.4. □

The neutrality result follows directly from the two sides of the model. The foreign economy must export tradables to acquire home money; the quantity it exports is determined by its need for the transaction asset, not by the tariff wedge. Formally, foreign steady-state behavior is governed solely by the service-market clearing condition (15), which does not contain θ. This pins down n^* (and hence K_F) independently of the tariff. At home, the tariff raises the consumer price of imported goods relative to the producer price, but—because tariff revenue is rebated lump-sum to home old-age consumers—the extra spending power exactly offsets the price increase. The home labor market therefore sees no change in the real return to tradables production, leaving n unchanged.

The tariff cannot eliminate the trade deficit because the deficit is the equilibrium counterpart of foreign demand for the home transaction asset, not an outcome of distorted prices.



(a) Foreign economy.

(b) Home economy.

Figure 4: Steady-state effects of an import tariff. Colors: the blue curve is labor supply (sectoral choice), the red curve is labor demand, and the dashed gray curve is the closed-economy labor demand with the gray dot denoting the closed-economy equilibrium. In the home panel, the purple dashed curve is labor demand in the tariff economy without the lump-sum tariff rebate. Despite shifts in home labor demand, the equilibrium mobile labor allocation n is unchanged relative to the no-tariff open-economy benchmark.

3.1 A labor supply–labor demand view of tariffs

In steady state, the foreign economy is unaffected by the home import tariff: the foreign labor supply schedule is unchanged and the foreign labor demand condition continues to pin down n^* as in Proposition 1. In the home economy, the tariff modifies the relationship between relative prices and expenditure shares (because tariff revenue is rebated lump-sum to home old-age consumers). Combining steady-state service demand with $p^k = (1+\theta)p^{k*}$ yields the home "labor demand" schedule under a tariff:

$$\frac{p^k}{p^c} = \frac{\phi}{1-\phi} \frac{(1-n)^\alpha}{\left(1-\frac{1}{\mu}\right)\frac{1}{1+\theta} n^{*\,\alpha-1} + \frac{\theta}{1+\theta}\left(n^{*\,\alpha} - \frac{\phi}{\mu} n^{*\,\alpha-1}\right) + n^\alpha} \qquad \text{(steady state).}$$

Figure 4 illustrates the key point: although the tariff shifts the home labor-demand schedule, the intersection with the unchanged labor-supply schedule occurs at the same n. The tariff therefore does not "reindustrialize" the home economy in the long run.

3.2 Valuation Effects and Trade-Deficit Measurement

What the tariff does change is relative prices. Home-produced goods are always more expensive than foreign-produced goods by the factor $(1+\theta)$. A higher tariff therefore compresses the foreign producer price relative to the home consumer price. Since import quantities are unchanged, this price wedge is the only channel through which the measured trade deficit can move.

Denote foreign demand for foreign-produced goods by K^*_{Ft}. Then

$$K^*_{Ft} = \phi \frac{p^{k*}_{t-1}}{p^{k*}_t} n^{*\alpha-1}_{t-1}$$

and the amount of goods imported by the home economy is

$$K_{Ft} = n^{*\alpha}_t - K^*_{Ft} = n^{*\alpha}_t - \phi \frac{p^{k*}_{t-1}}{p^{k*}_t} n^{*\alpha-1}_{t-1}$$

In the benchmark model, home exports are zero, so the trade deficit equals import spending valued at world prices. Under tariffs, three accounting concepts diverge: imports valued at world prices ($p^{k*}K_F$), imports valued at tariff-inclusive domestic prices ($p^k K_F = (1+\theta)p^{k*}K_F$), and domestic expenditure (which now includes tariff revenue). The measured trade-deficit-to-GDP ratio uses world-price imports in the numerator and domestic-price GDP in the denominator:

$$\frac{\mathrm{EXP}_t}{\mathrm{GDP}_t} - 1 = \frac{p^{k*}_t K_{Ft}}{p^k_t n^\alpha_t + p^c_t (1-n_t)^\alpha}.$$

The trade deficit as a share of GDP is

$$\frac{\mathrm{EXP}_t}{\mathrm{GDP}_t} - 1 = \frac{p^{k*}_t}{p^k_t} \frac{n^{*\alpha}_t - \phi \frac{p^{k*}_{t-1}}{p^{k*}_t} n^{*\alpha-1}_{t-1}}{n^{\alpha-1}_t}$$

In steady state (after substituting for n^*):

$$\frac{\mathrm{EXP}_t}{\mathrm{GDP}_t} - 1 = \frac{1}{1+\theta} \left(\frac{\mu - 1}{\mu} \right) n^{*\alpha-1} \frac{n}{n^\alpha}.$$

The measured trade deficit falls as the tariff rises. But the decline is entirely mechanical. The import quantity K_F is pinned down by foreign production and demand—by μ, ϕ, and α—and is invariant to θ. What changes is the price at which imports enter the trade-balance calculation: a higher tariff depresses $p^{k*}/p^k = 1/(1+\theta)$, and since imports are valued at world prices while home GDP is valued at domestic prices, the ratio falls. This is a valuation effect, not a quantity effect. The home economy is not importing less; it is simply valuing its imports more cheaply

relative to its own output.

3.3 Burden Sharing and Incidence

The long-run picture is this: quantities are unchanged, but prices are not. The tariff drives a wedge between what home consumers pay and what foreign producers receive. Both sides of this wedge are borne by someone.

Foreign exporters receive lower net-of-tariff prices—a terms-of-trade deterioration. Home consumers pay more for imported tradables. The home government collects the difference. In this sense the tariff looks more like an international income transfer than a border protection device: it shifts resources from foreign producers and home consumers to the home fiscal authority, without altering the real quantities that cross the border.

To see why incidence is shared, note that in equilibrium $p^k = (1+\theta)p^{k*}$. A higher tariff either raises the home consumer price p^k, compresses the foreign producer price p^{k*}, or some combination of both. In our model, both adjust, so both home consumers and foreign producers bear part of the wedge. This shared-incidence pattern aligns with the empirical findings in Amiti et al. (2019) and Fajgelbaum et al. (2020), and with the broader message on valuation channels in Caliendo et al. (2025) and Itskhoki and Mukhin (2025).

Finally, the neutrality result does not depend on the Cobb–Douglas (unit-elasticity) specification for preferences over goods and services. Appendix C replaces the Cobb–Douglas aggregator with a general CES aggregator with elasticity $\varepsilon > 0$ and shows that tariff neutrality continues to hold: the tariff rate drops out of the steady-state equilibrium conditions through the same cancellation mechanism as in the benchmark (Proposition 9).

4 Domestic Industrial Policy

Suppose that, instead of imposing a tariff, the home country subsidizes production in the goods-producing (tradables) sector, financed by a lump-sum tax. The decision problem in the foreign country is unchanged.

Home producers in the goods-producing sector receive a subsidy κ per unit of output sold. Since the subsidy is paid to producers rather than levied at the border, the law of one price continues to hold for consumer prices: $p_t^k = p_t^{k*}$. Therefore,

$$
\begin{aligned}
w_t^1 &= (1+\kappa)\, p_t^k\, (1-\alpha)\, n_t^{\alpha} \\
w_t^2 &= p_t^c\, (1-\alpha)\, (1-n_t)^{\alpha} \\
w_t^3 &= (1+\kappa)\, p_t^k \alpha n_t^{\alpha-1} = p_t^c \alpha\, (1-n_t)^{\alpha-1}
\end{aligned}
$$

Therefore,

$$\frac{p_t^c}{p_t^k} = (1+\kappa)\, n_t^{\alpha-1} \frac{1-n_t}{(1-n_t)^{\alpha}} = (1+\kappa)\, \frac{n_t^{\alpha-1}}{(1-n_t)^{\alpha-1}} \tag{18}$$

The government budget constraint in the home economy is

$$M_t - M_{t-1} - \kappa p_t^k n_t^{\alpha} = \tau_t.$$

The demand in the home economy is

$$\begin{aligned} p_{t+1}^c c_{t+1} &= (1-\phi)\left(\mu\,(1+\kappa)\, p_t^k n_t^{\alpha-1} + (\mu-1)\, p_t^{k*} n_t^{*\alpha-1} - \kappa p_{t+1}^k n_{t+1}^{\alpha}\right) \\ p_{t+1}^k K_{Ht+1} + p_{t+1}^{k*} K_{Ft+1} &= \phi\left(\mu\,(1+\kappa)\, p_t^k n_t^{\alpha-1} + (\mu-1)\, p_t^{k*} n_t^{*\alpha-1} - \kappa p_{t+1}^k n_{t+1}^{\alpha}\right) \end{aligned} \tag{19}$$

and in the foreign economy

$$\begin{aligned} p_{t+1}^{c*} c_{t+1}^* &= (1-\phi)\, p_t^{k*} n_t^{*\alpha-1} \\ p_{t+1}^{k*} K_{Ft+1}^* &= \phi p_t^{k*} n_t^{*\alpha-1} \end{aligned} \tag{20}$$

along with optimal sector choice condition (22).

Imposing steady state in equation (20) and using optimal sectoral condition (22):

$$\begin{aligned} n^* &= \frac{\mu-1}{\mu} + \frac{\phi}{\mu} \\ k_F^* &= \frac{\phi}{\mu} n^{*\alpha-1} \end{aligned}$$

and from market clearing, imports in the home country are

$$K_F = n^{*\alpha} - \frac{\phi}{\mu} n^{*\alpha-1} = \left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1}$$

Rewriting equation (19) in steady state and eliminating relative prices using equation (18):

$$(1+\kappa)\left(n^{\alpha-1} - n^{\alpha}\right) = (1-\phi)\left((1+\kappa)\, n^{\alpha-1} + \left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1} - \kappa n^{\alpha}\right)$$

which yields

$$(1+\kappa)\,\phi n^{\alpha-1} - (1+\phi\kappa)\, n^{\alpha} = (1-\phi)\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1}.$$

Unlike in the benchmark and the tariff case, the home labor allocation depends on κ: as subsidies increase, employment and output in the goods-producing sector increase.

Proposition 3 (Industrial Policy)**.** *Suppose $\kappa \geq 0$ is a production subsidy to home tradables. In steady state:*

(i) The foreign labor allocation n^* and the import quantity K_F are invariant to κ.

(ii) The home tradables labor share n is strictly increasing in κ.

Proof. See Appendix A.6. □

The production subsidy succeeds because it operates on a different margin from the tariff. A tariff acts on the *border price of the imported good* and, because the revenue is rebated, has no net effect on the relative return to home tradables production. A production subsidy, by contrast, directly raises the *marginal revenue product of labor in the home goods sector*, shifting the labor supply curve for mobile workers in favor of tradables. Foreign workers face exactly the same conditions as before, so n^* and K_F are unaffected. The takeaway is that the channel through which policy raises n matters: border-price instruments (tariffs) that merely redistribute income between sectors cannot achieve what production-side instruments (subsidies) that alter the private return to production can.

Graphical interpretation. The subsidy directly raises the marginal revenue product of labor in the home goods sector. In the labor-supply/labor-demand diagram, this shows up as a shift in the home labor supply schedule (sectoral choice) for the mobile factor, while the demand schedule is unchanged. Figure 5 illustrates how a production subsidy raises the steady-state mobile labor allocation n in the home tradables sector.

The subsidy does not affect the quantity of goods imported. But because the relative wage in the goods sector rises, home goods production increases and home services production falls. The trade deficit as a share of GDP is

$$\begin{aligned}\frac{p_t^{k*} K_{Ft}}{p_t^k n_t^\alpha + p_t^c (1-n_t)^\alpha} &= \frac{p_t^{k*}}{p_t^k} \frac{\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1}}{n^\alpha + \frac{p_t^c}{p_t^k}(1-n_t)^\alpha} \\ &= \frac{\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1}}{(1+\kappa)\, n^{\alpha-1} - \kappa n^\alpha}\end{aligned}$$

The denominator can be rewritten as

$$(1+\kappa)\, n^{\alpha-1} - \kappa n^\alpha = \left(\frac{1-\phi}{\phi}\right)\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1} + \frac{n^\alpha}{\phi}$$

which is increasing in κ. So, as with the tariff, the subsidy reduces the trade-deficit-to-GDP ratio, despite having no effect on the import quantity. The difference is that under the subsidy, more

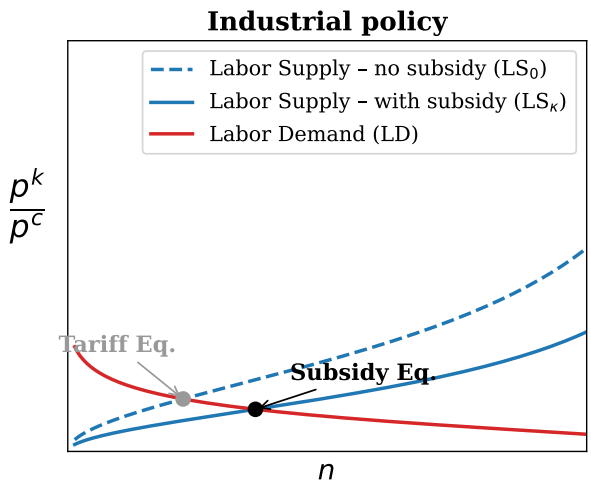


Figure 5: Home-economy steady state with a production subsidy to the goods sector. Colors: the red curve is labor demand, the solid blue curve is labor supply with the subsidy, and the dashed blue curve is labor supply without the subsidy; the gray dot denotes the equilibrium without the subsidy. The subsidy shifts the home labor-supply schedule up, raising the equilibrium mobile labor share in the goods sector n relative to the no-subsidy open-economy benchmark.

goods and fewer services are produced at home, and households consume relatively more goods than in the benchmark.

Tariffs and production subsidies are both government interventions in the tradables sector, but they operate on different margins. If the goal is reindustrialization, the instrument has to reach the right margin: the relative return to producing tradables at home.

5 Extensions

The benchmark results rest on several simplifications worth relaxing. First, we assumed home and foreign tradables are perfect substitutes, so demand elasticity played no role. Second, we assumed tariff revenue is rebated lump-sum; alternative fiscal arrangements may modify the neutrality logic. Third, the benchmark features a single transaction asset with no competing store of value. We take up each in turn.

5.1 Imperfect substitution between home and foreign tradables

Consider an extension in which home and foreign goods are imperfect substitutes. Both countries have CES preferences over goods varieties. Throughout this section, lowercase letters (k_H, k_F, etc.) denote individual demands, while aggregate quantities are denoted by uppercase letters or, where individual and aggregate coincide (one agent per generation), by the same symbol.

Preferences in the home economy are given by

$$u\left(k_H, k_F, c\right) = \left[(1-\eta)^{\frac{1}{\rho}} k_H^{\frac{\rho-1}{\rho}} + \eta^{\frac{1}{\rho}} k_F^{\frac{\rho-1}{\rho}}\right]^{\frac{\phi\rho}{\rho-1}} c^{1-\phi}, \quad 0 < \rho < \infty,$$

where c is consumption of services, k_H is consumption of home-produced goods, and k_F is consumption of foreign-produced goods. The parameter η determines the degree of home bias in goods consumption in the home economy. Similarly, preferences in the foreign economy are given by

$$u\left(k_H^*, k_F^*, c^*\right) = \left[\eta^{*\frac{1}{\gamma}} k_H^{*\frac{\gamma-1}{\gamma}} + (1-\eta^*)^{\frac{1}{\gamma}} k_F^{*\frac{\gamma-1}{\gamma}}\right]^{\frac{\phi\gamma}{\gamma-1}} c^{*1-\phi}, \quad 0 < \gamma < \infty,$$

where c^* is consumption of services, k_H^* is consumption of home-produced goods by foreign consumers, and k_F^* is consumption of foreign-produced goods by foreign consumers. The parameter η^* determines the degree of home bias in goods consumption in the foreign economy. The benchmark economy studied earlier corresponds to $\gamma, \rho \to \infty$.

As before, let $\left(p^k, p^c\right)$ and $\left(p^{k*}, p^{c*}\right)$ be the prices of goods and services produced in the home and foreign economies, respectively. Let $q \equiv p^k/p^{k*}$ denote the relative producer price of the home variety. The home country levies an ad valorem tariff $\theta \geq 0$ on the foreign variety, so the tariff-inclusive home import price is $(1+\theta)\, p^{k*}$.

The definition of equilibrium is straightforward and similar to the case where home and foreign goods are perfect substitutes. Producer optimality together with optimal sector choice by mobile workers in the home and foreign economies implies

$$\frac{p^k}{p^c} = \left(\frac{1-n}{n}\right)^{\alpha-1}, \tag{21}$$

$$\frac{p^{k*}}{p^{c*}} = \left(\frac{1-n^*}{n^*}\right)^{\alpha-1}. \tag{22}$$

The demands by old workers for goods and services in the foreign economy are given by

$$p_{t+1}^{c*} c_{t+1}^* = (1-\phi)\, p_t^{k*} n_t^{*\alpha-1} \tag{23}$$

$$p_{t+1}^{k*} k_{Ft+1}^* = \frac{1-\eta^*}{1-\eta^* + \eta^* \left(\frac{p_{t+1}^k}{p_{t+1}^{k*}}\right)^{1-\gamma}} \phi p_t^{k*} n_t^{*\alpha-1}$$

$$k_{Ht+1}^* = \frac{\eta^*}{1-\eta^*} \left(\frac{p_{t+1}^k}{p_{t+1}^{k*}}\right)^{-\gamma} k_{Ft+1}^* \tag{24}$$

where we replaced asset holdings using $M_t^* = p_t^{k*} n_t^{*\alpha} + p_t^{c*} (1 - n_t^*)^\alpha = p_t^{k*} n_t^{*\alpha-1}$. The first equality states that money holdings in period t by young workers must equal GDP, and the second equality follows from optimal sectoral choice. Similarly, the demands by old workers for goods and services in the home economy are given by

$$
\begin{aligned}
p_{t+1}^c c_{t+1} &= (1-\phi)\left(\mu p_t^k n_t^{\alpha-1} + (\mu-1) p_t^{k*} n_t^{*\alpha-1} + \theta p_{t+1}^{k*} k_{Ft+1}\right) \\
(1+\theta) p_{t+1}^{k*} k_{Ft+1} &= \frac{\eta \left((1+\theta)\frac{p_{t+1}^{k*}}{p_{t+1}^k}\right)^{1-\rho}}{1-\eta+\eta\left((1+\theta)\frac{p_{t+1}^{k*}}{p_{t+1}^k}\right)^{1-\rho}} \phi \left(\mu p_t^k n_t^{\alpha-1} + (\mu-1) p_t^{k*} n_t^{*\alpha-1} + \theta p_{t+1}^{k*} k_{Ft+1}\right) \\
k_{Ht+1} &= \frac{1-\eta}{\eta}\left((1+\theta)\frac{p_{t+1}^{k*}}{p_{t+1}^k}\right)^{\rho} k_{Ft+1}
\end{aligned}
$$

Note that old workers spend their money holdings, M_t, as well as the transfers funded through seigniorage $(\mu-1)(M_t + M_t^*)$ and tariff revenue $\theta p_{t+1}^{k*} k_{Ft+1}$. We simplified this sum using the income accounting identity $M_t = p_t^k n_t^\alpha + p_t^c (1-n_t)^\alpha$ and the optimal sectoral choice conditions described above. Finally, the equilibrium allocations must satisfy market clearing for home and foreign produced goods,

$$k_{Ht} + k_{Ht}^* = n_t^\alpha, \quad k_{Ft} + k_{Ft}^* = n_t^{*\alpha},$$

and for services,

$$c_t = (1-n_t)^\alpha, \quad c_t^* = (1-n_t^*)^\alpha.$$

In equilibrium the following balance-of-payments identity holds:

$$p_t^{k*} k_{Ft} - p_t^k k_{Ht}^* = M_t^* - M_{t-1}^*. \tag{25}$$

Equation (25) states that the value of new money acquired by the foreign economy must equal the value of the home country's trade deficit in goods.

In the steady state all allocations and relative prices are constant and all price levels grow at rate μ. Using equation (23), imposing market clearing for services in the foreign economy in steady state, and using optimal sectoral choice to replace p^{k*}/p^{c*}, we can solve for the labor allocation n^*:

$$n^* = \frac{\mu-1}{\mu} + \frac{\phi}{\mu}.$$

Note that the labor allocation in the foreign economy is identical to the benchmark case and is independent of the tariff rate θ.

Next we write demand for goods in the foreign economy in steady state:

$$k_F^* = \frac{1-\eta^*}{1-\eta^*+\eta^* q^{1-\gamma}} \frac{\phi}{\mu} n^{*\alpha-1},$$
$$k_H^* = \frac{\eta^*}{1-\eta^*} q^{-\gamma} k_F^*.$$

Note that demand for goods in the foreign economy does not directly depend on the tariff rate. It only depends on the tariff through dependence on q, the relative producer price of the home variety. Moreover, foreign demand for the foreign goods variety, k_F^*, is increasing in q, and foreign demand for the home goods variety, k_H^*, is decreasing in q. Market clearing for foreign-produced goods implies that, in equilibrium, home demand for the foreign goods variety, $k_F = n^{*\alpha} - k_F^*$, is decreasing in q. Moreover, it only depends on the tariff rate indirectly, through its dependence on q.

Lemma 3. *In the steady-state equilibrium,* $\partial k_H^*/\partial q < 0$, $\partial k_F/\partial q < 0$, *and* $\partial k_F^*/\partial q > 0$.

Note that in steady state the balance-of-payments equation (25) can be written as

$$k_F - q k_H^* = \left(1 - \frac{1}{\mu}\right) n^{*\alpha-1}.$$

This immediately implies that the real net trade deficit is invariant to the tariff rate θ. However, the flow of goods k_F and k_H^* can change. Below, we show that the steady-state values of these gross trade flows decrease with the tariff rate. In order to establish this claim, we need to show that the steady-state equilibrium relative price q is increasing in the tariff rate θ. We start by writing demand for goods and services in the home economy in steady state, after imposing market clearing and using optimal sectoral choice to simplify the equations:

$$\phi n^{\alpha-1} - n^{\alpha} = (1-\phi)\left(\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1} + \theta k_F\right)\frac{1}{q} \tag{26}$$
$$k_F = \frac{\eta\left(\frac{1+\theta}{q}\right)^{1-\rho}}{1-\eta+\eta\left(\frac{1+\theta}{q}\right)^{1-\rho}} \phi\left(\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1} + \theta k_F\right)\frac{1}{q}$$
$$k_H = \frac{1-\eta}{\eta}\left(\frac{1+\theta}{q}\right)^{\rho} k_F \tag{27}$$

Lemma 4. *In the steady-state equilibrium,* $\partial q/\partial\theta > 0$.

Proof. We start from equation (27). Differentiating:

$$\frac{\partial k_H}{\partial \theta} = \rho \frac{k_H}{1+\theta} + \left(\frac{k_H}{k_F} \frac{\partial k_F}{\partial q} - \rho \frac{k_H}{q} \right) \frac{\partial q}{\partial \theta}. \tag{28}$$

Using market clearing for home-produced goods, $k_H + k_H^* = n^\alpha$:

$$\frac{\partial k_H}{\partial \theta} + \frac{\partial k_H^*}{\partial q} \frac{\partial q}{\partial \theta} = \alpha n^{\alpha-1} \frac{\partial n}{\partial \theta}.$$

Now replace $\partial k_H / \partial \theta$ from (28):

$$\rho \frac{k_H}{1+\theta} + \left(\frac{k_H}{k_F} \frac{\partial k_F}{\partial q} - \rho \frac{k_H}{q} + \frac{\partial k_H^*}{\partial q} \right) \frac{\partial q}{\partial \theta} = \alpha n^{\alpha-1} \frac{\partial n}{\partial \theta}. \tag{29}$$

Next, we use demand for home services, equation (26):

$$\phi n^{\alpha-1} - n^\alpha = (1-\phi) \left(\left(\frac{\mu-1}{\mu} \right) \frac{1}{q} n^{*\alpha-1} + \theta \frac{1}{q} k_F \right).$$

Differentiating with respect to θ:

$$\left(\phi (\alpha-1) n^{\alpha-2} - \alpha n^{\alpha-1} \right) \frac{\partial n}{\partial \theta} = (1-\phi) \left(\left(\frac{\mu-1}{\mu} n^{*\alpha-1} + \theta k_F \right) \frac{-1}{q} + \theta \frac{1}{q} \frac{\partial k_F}{\partial q} \right) \frac{\partial q}{\partial \theta} + (1-\phi) \frac{1}{q} k_F.$$

Replacing $\partial n / \partial \theta$ from (29) and rearranging, $\partial q / \partial \theta = \mathcal{N} / \mathcal{D} > 0$, where

$$\begin{aligned} \mathcal{N} &= (1-\phi) \frac{k_F}{q} - \left(\phi \left(\frac{\alpha-1}{\alpha} \right) n^{-1} - 1 \right) \rho \frac{k_H}{1+\theta}, \\ \mathcal{D} &= \left(\phi \tfrac{\alpha-1}{\alpha} n^{-1} - 1 \right) \left[\frac{k_H}{k_F} \frac{\partial k_F}{\partial q} - \rho \frac{k_H}{q} + \frac{\partial k_H^*}{\partial q} \right] \\ &\quad - (1-\phi) \left(\left(\tfrac{\mu-1}{\mu} n^{*\alpha-1} + \theta k_F \right) \frac{-1}{q} + \frac{\theta}{q} \frac{\partial k_F}{\partial q} \right). \end{aligned}$$

Since $\alpha < 1$, the numerator is always positive. The denominator is also always positive, given Lemma 3. □

Together, Lemmas 3 and 4 imply that the home economy's imports of goods k_F and exports of goods k_H^* fall with the tariff rate θ in the steady-state equilibrium.

Proposition 4 (Gross Trade Compression)**.** *In the steady-state equilibrium, the home economy's imports of goods k_F and exports of goods k_H^* fall with the tariff rate θ.*

Next, we examine how the allocation of mobile workers in the goods-producing sector of the home economy, n, varies with the tariff rate θ. We show that for any positive tariff rate, the allocation n is weakly lower relative to the case where there is no tariff.

Proposition 5 (Home Tradables Employment)**.** *Let $n(\theta)$ and $q(\theta)$ be the steady-state equilibrium relative price and allocation of mobile workers in the goods-producing sector of the home economy as a function of the tariff rate θ. Then $n(0) \geq n(\theta)$ for any $\theta > 0$.*

Proof. See Appendix A.5. □

What does this imply for the measured trade deficit as a share of GDP in the home economy in steady state? Recall that GDP in the home economy is $p^k n^\alpha + p^c (1-n)^\alpha = p^k n^{\alpha-1}$.

$$\frac{\text{EXP}}{\text{GDP}} - 1 = \frac{p^{k*} k_F - p^k k_H^*}{p^k n^{\alpha-1}} = \frac{k_F - q k_H^*}{q n^{\alpha-1}} = \frac{\left(1 - \frac{1}{\mu}\right) n^{*\alpha-1}}{q n^{\alpha-1}}.$$

Note that $n(\theta)^{\alpha-1} > n(0)^{\alpha-1}$ for any $\theta > 0$. Therefore, the measured trade deficit as a share of GDP is lower when the tariff is positive relative to the equilibrium with no tariff.

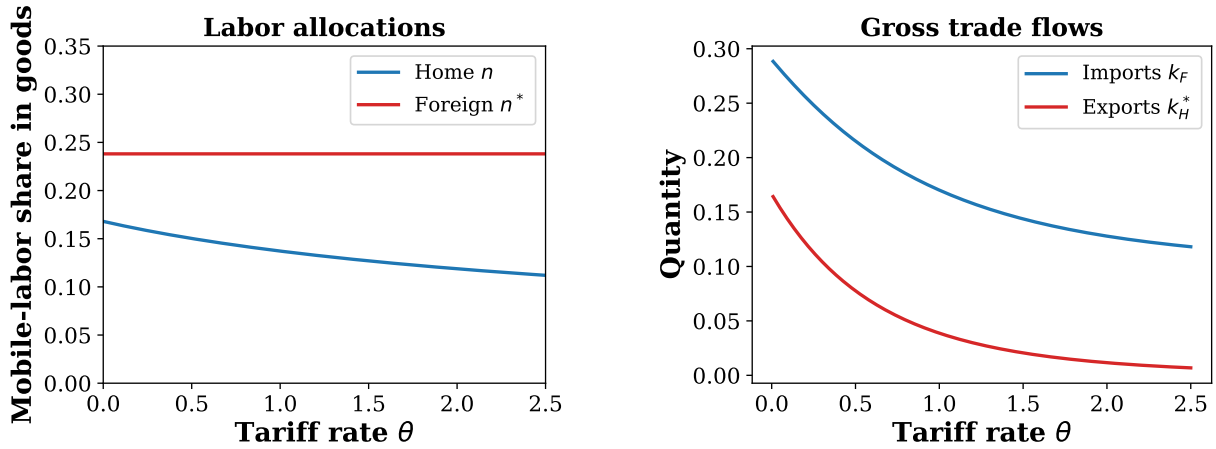
In this setting, where both countries substitute between home and foreign tradables, the reserve-currency mechanism still pins down the net real resource transfer, but tariffs can compress gross trade flows and increase the relative producer price of home-produced goods. Moreover, tariffs further shift home production toward nontradables relative to a world where there is no tariff. The key takeaway is that relaxing the perfect substitutability assumption does not overturn the benchmark result: the tariff still does not undo the reserve-demand mechanism that governs the net transfer.

Figure 6 reports a numerical comparative statics exercise. As the tariff increases: (i) the foreign goods-sector labor share is essentially unchanged; (ii) the home goods-sector labor share declines (further deindustrialization); and (iii) both gross trade flows fall.

Below, we show that in a special case the exact tariff neutrality in the long run holds even when foreign and home produced goods are not perfect substitutes.

A special case: perfect substitution in foreign economy

The results above hold for any CES preferences in the home and foreign economies. Here we show that if home and foreign goods are perfect substitutes in the foreign economy, $\gamma \to \infty$, then the full tariff neutrality result in the long run is restored. To see this, note that with $\gamma \to \infty$, equation (24) implies that demand for home-produced goods in the foreign economy collapses to zero. Therefore, similar to the benchmark model, the trade deficit in the home economy equals



(a) Labor allocations.

(b) Gross trade flows.

Figure 6: Symmetric CES comparative statics ($\phi = 0.2$, $\alpha = 0.5$, $\mu = 1.05$, $\rho = \gamma = 4$, $\eta = \eta^* = 0.6$). As the import tariff θ increases, the home goods-sector labor share n declines (further deindustrialization, left panel), while the foreign goods-sector labor share n^* remains unchanged. Both gross trade flows—imports k_F and exports k_H^*—fall monotonically with θ (right panel).

the value of home imports of foreign-produced goods. The balance of payments (25) in steady state implies that

$$k_F = \left(1 - \frac{1}{\mu}\right) n^{*\alpha-1}.$$

In other words, both the net balance of trade and gross trade flows are invariant to the tariff rate in the long run.

On the other hand, market clearing for home-produced goods implies $k_H = n^\alpha$. Replacing these into equation (27):

$$n^\alpha = \frac{1-\eta}{\eta}\left(\frac{1+\theta}{q}\right)^\rho \left(1 - \frac{1}{\mu}\right) n^{*\alpha-1}. \tag{30}$$

On the other hand, we can replace k_F in equation (26) and rewrite as

$$\phi n^{\alpha-1} - n^\alpha = (1-\phi)\left(\frac{1+\theta}{q}\right)\left(\frac{\mu-1}{\mu}\right) n^{*\alpha-1}. \tag{31}$$

Eliminating $(1+\theta)/q$ between (30) and (31) shows that the steady-state labor allocation n is independent of the tariff rate θ. Long-run sectoral allocations and outputs are invariant to the tariff even when tradables are imperfect substitutes at home. Note, however, that the relative price increases linearly with $1+\theta$.

The trade deficit as a share of home GDP in steady state is

$$\frac{\text{EXP}}{\text{GDP}} - 1 = \frac{p^{k*} k_F}{p^k n^{\alpha-1}} = \left(1 - \frac{1}{\mu}\right) \frac{n^{*\alpha-1}}{q n^{\alpha-1}},$$

which is decreasing in θ through the price ratio q. As in the benchmark, the measured deficit falls through valuation rather than through quantity adjustment.

Proposition 6 (CES Tariff Neutrality)**.** *Under one-way CES demand (home consumers substitute between home and foreign tradables with elasticity $\rho > 0$; foreign consumers are specialized in the foreign variety), the steady-state labor allocations (n, n^*) and the import quantity k_F are invariant to the tariff rate θ. The measured trade-deficit-to-GDP ratio falls with θ through the relative price q, not through any change in quantities.*

5.2 Alternative Fiscal Policy

The benchmark analysis assumes tariff revenue is rebated lump-sum to home old-age consumers, leaving μ exogenous. Here we consider an alternative fiscal arrangement in which the home government has an exogenous expenditure requirement $g \in (0, 1)$: it must consume a fraction g of nominal GDP each period. The government finances this expenditure with seigniorage *and* tariff revenue combined (government budget constraint (33)). The money-growth rate μ is endogenous, but the tariff enters the government budget constraint in a way that leaves the steady-state conditions for n and n^* unaffected. The full formal definitions, equilibrium conditions, and steady-state derivations are in Appendix B.

Proposition 7 (Tariff Neutrality with Fiscal Spending)**.** *The steady-state labor allocations n and n^* are invariant to the tariff rate θ. There exists a unique inflation rate $\mu \in \left(1, \frac{1}{1-g}\right)$ satisfying the government budget constraint, independently of θ.*

Proof. See Appendix A.7. □

After imposing steady state, n and n^* are pinned down solely by μ (as in the benchmark), and the budget constraint determines μ independently of θ. Intuitively, tariff revenue substitutes for seigniorage in financing g, but since allocations depend only on μ—not on the composition of government revenue—this substitution leaves the equilibrium unchanged.

5.3 Robustness: Multi-Country Environment with Portfolio Choice

The benchmark model assumes that the home transaction asset is the only store of value and that the foreign economy is a single country. Appendix D relaxes both assumptions by developing a

multi-country environment in which a continuum of foreign economies produce differentiated goods (Dixit-Stiglitz varieties), trade with each other using the home currency for cross-border settlement, and allocate savings between home money and a competing foreign settlement instrument. The portfolio share between the two currencies is determined endogenously by a micro-founded transaction problem: each cross-border purchase requires currency, and different markets have different acceptance rates for the reserve currency, generating a CES liquidity aggregator from primitives.

The central result is that tariff neutrality survives in this richer environment (Proposition 10 in Appendix D). The logic is the same as in the benchmark, but now applies to a setting with endogenous portfolio choice. The import tariff θ creates a wedge in goods prices, but the portfolio share s—which determines the net real transfer and thus the trade deficit—depends on asset returns $(1/\mu, 1/\hat{\mu})$ and the structure of international payment networks (λ, σ), not on goods prices. Because the tariff operates on the goods-price margin while the trade deficit is pinned by the asset-return margin, the two are orthogonal: the tariff cannot alter the portfolio share, the foreign labor allocation, or the import quantity.

This orthogonality does not depend on the specific functional form of the portfolio problem. As shown in Appendix D, the neutrality holds for any smooth, homogeneous-of-degree-one liquidity aggregator $L(a, b)$, not just CES. The key requirement is only that the tariff does not enter the asset-return condition that determines the portfolio share—a property that holds whenever the tariff is a pure goods-price wedge and revenue is rebated to domestic agents.

The appendix also contrasts the goods tariff with a *financial tariff* τ_f on foreign reserve holdings. Because the financial tariff reduces the after-tax return on the home asset, it operates directly on the asset-return margin: it shifts the portfolio share, reduces the trade deficit, and expands home tradables employment. The contrast between the neutral goods tariff and the effective financial tariff sharpens the paper's main message: in a reserve-currency economy, the relevant instrument for external rebalancing is financial policy, not trade policy.

6 Conclusion

If the U.S. trade deficit is the equilibrium counterpart of foreign demand for dollar-denominated assets, then the deficit cannot be eliminated by making imports more expensive. The foreign economy exports tradables not because it wants to, but because it needs dollars. A tariff does not change that need.

We have built a model to make this argument precise. In equilibrium, the foreign labor allocation and the import quantity are governed entirely by the foreign economy's demand for the home transaction asset—both are independent of the tariff rate. At home, the tariff raises import

prices, but when revenue is rebated to consumers their spending power is restored. The home labor market sees no change in the relative return to tradables production. Sectoral employment and output are invariant.

What does change is relative prices. A higher tariff compresses the foreign producer price relative to the home consumer price, reducing the measured trade-deficit-to-GDP ratio through valuation rather than through quantity adjustment. The burden is shared: foreign exporters receive less per unit, home consumers pay more per unit, and the home government collects the difference.

Several extensions refine this message. Imperfect substitutability between home and foreign tradables does not restore tariff effectiveness. In the special case where foreign consumers are specialized in the foreign variety, long-run quantities remain fully invariant and only the valuation channel operates (Proposition 6). In the general symmetric CES case, where both countries substitute between varieties, the net real resource transfer is still pinned down by reserve demand, but tariffs compress gross trade flows (Proposition 4) and further reduce home tradables employment (Proposition 5). A production subsidy to home tradables does raise steady-state employment in the goods sector, because it operates directly on the margin that tariffs miss: the relative return to producing tradables (Proposition 3). The fiscal disposition of tariff revenue matters: if revenue finances a fixed government expenditure rather than being rebated to consumers, neutrality continues to hold (Proposition 7). Most importantly, the neutrality result is not an artifact of the benchmark's single-currency assumption: it survives in a multi-country environment with differentiated goods, a foreign settlement currency, and endogenous portfolio choice (Proposition 10). In that richer environment, a financial tariff on foreign reserve holdings—unlike a goods tariff—does reduce the trade deficit and expand home tradables employment, because it operates on the asset-return margin rather than the goods-price margin (Proposition 11).

The broader lesson is about the interaction between trade policy and the monetary system. In a reserve-currency economy, the real trade balance is determined by asset demand, not by trade policy. Tariffs can redistribute income between countries and raise domestic revenue; they cannot rebalance trade or reindustrialize the economy on their own. A policy aimed at reducing the trade deficit in real terms would need to address the underlying demand for U.S.-issued assets—a much harder problem than adjusting the tariff schedule.

Several important dimensions are left for future work. The model abstracts from aggregate demand, nominal rigidities, and short-run exchange-rate dynamics, all of which may give tariffs non-trivial transitional effects even when long-run allocations are neutral. Introducing a discrete erosion of reserve-currency status as a response to high tariffs would add a strategic dimension not captured here. These extensions are left for future research.

A Proofs

A.1 Proof of Lemma 1

Start from the demand for services in the foreign country

$$p_t^{c*} \sum_{i=1,2,3} c_t^{i*} = (1-\phi) \sum_{i=1,2,3} w_{t-1}^{i*}$$

On the left-hand side, total consumption of services equals total production. On the right-hand side, total income in the foreign economy equals total value of output. Therefore,

$$p_t^{c*} F(1-n_t^*) = (1-\phi) \left(p_{t-1}^k n_{t-1}^{*\alpha} + p_{t-1}^{c*} \left(1-n_{t-1}^*\right)^\alpha \right)$$

The optimal choice by the type 3 worker implies

$$\frac{p_t^k}{p_t^{c*}} = \frac{n_t^*}{1-n_t^*} \frac{(1-n_t^*)^\alpha}{n_t^{*\alpha}}$$

Using this to simplify gives equation (6):

$$\left(1-n_t^*\right) n_t^{*\alpha-1} = (1-\phi) \frac{p_{t-1}^k}{p_t^k} n_{t-1}^{*\alpha-1}$$

Similarly, demand for goods in the foreign economy is

$$\sum_{i=1,2,3} k_t^{i*} = \phi \frac{p_{t-1}^k}{p_t^k} n_{t-1}^{*\alpha-1}$$

The demand for service in the home economy is

$$p_t^c \sum_{i=1,2,3} c_t^i = (1-\phi)(\sum_{i=1,2,3} w_{t-1}^i + (\mu-1) M_{t-1})$$

where

$$\sum_{i=1,2,3} w_{t-1}^i = p_{t-1}^k n_{t-1}^\alpha + p_{t-1}^c \left(1-n_{t-1}\right)^\alpha$$

and

$$M_{t-1} = p_{t-1}^k \left(n_{t-1}^\alpha + n_{t-1}^{*\alpha} \right) + p_{t-1}^{c*} \left(1-n_{t-1}^*\right)^\alpha + p_{t-1}^c \left(1-n_{t-1}\right)^\alpha$$

The first equation is the national income accounting identity. The second is the total demand for money by the young generation of cohort $t-1$. The optimal sector choice of type 3 at home

implies

$$\frac{p_t^k}{p_t^c} = \frac{n_t}{1-n_t}\frac{(1-n_t)^\alpha}{n_t^\alpha}$$

Using these equations to simplify service demand at home gives equation (7):

$$(1-n_t)\, n_t^{\alpha-1} = (1-\phi)\,\frac{p_{t-1}^k}{p_t^k}\left(\mu n_{t-1}^{\alpha-1} + (\mu-1)\, n_{t-1}^{*\alpha-1}\right)$$

Similarly, demand for goods in the home economy is

$$k_t = \phi\frac{p_{t-1}^k}{p_t^k}\left(\mu n_{t-1}^{\alpha-1} + (\mu-1)\, n_{t-1}^{*\alpha-1}\right)$$

Adding demand for goods in home and foreign economies and using goods-market clearing gives equation (8):

$$n_t^\alpha + n_t^{*\alpha} = \phi\mu\frac{p_{t-1}^k}{p_t^k}\left(n_{t-1}^{\alpha-1} + n_{t-1}^{*\alpha-1}\right)$$

The initial old in the home economy hold the initial stock of money M_0 and can buy goods and services. Service demand at home together with service-market clearing implies equation (9):

$$p_0^c\,(1-n_0)^\alpha = (1-\phi)\, M_0.$$

The initial old in the foreign economy have no money and hence no consumption. Since foreign services are non-tradable and no one can pay for them, no services are produced. All labor goes to goods production, which is shipped to the home economy in exchange for money. Together with optimal goods demand in the domestic economy this implies equation (10):

$$p_0^k\,(n_0^{*\alpha} + n_0^\alpha) = \phi M_0.$$

Finally, equation (11) is the optimal sector choice for type 3 households in the home economy.

A.2 Proof of Proposition 1

Imposing a steady-state condition in (6) we get

$$1 - n^* = \frac{1-\phi}{\mu}$$

and therefore

$$n^* = 1 - \frac{1-\phi}{\mu}.$$

Therefore, $n^* \geq \phi$ with strict inequality if $\mu > 1$.

Next, imposing the steady-state condition in (7):

$$\frac{1-n}{n} n^{\alpha} = (1-\phi)\left(\frac{n^{\alpha}}{n} + \frac{\mu-1}{\mu}\frac{n^{*\alpha}}{n^*}\right)$$

Therefore, $n \leq \phi$ (since $\mu \geq 1$) with a strict inequality if $\mu > 1$. This establishes the first claim.

For the second claim, the ratio of the wage of a goods producer to that of a service producer in the home country is

$$\begin{aligned}\frac{w_t^1}{w_t^2} &= \frac{p_t^k}{p_t^c}\frac{n^{\alpha}}{(1-n)^{\alpha}} \\ &= \frac{n}{1-n} \\ &\leq \frac{\phi}{1-\phi},\end{aligned}$$

with equality if and only if $\mu = 1$. A similar argument establishes $\frac{w_t^{1*}}{w_t^{2*}} \geq \frac{\phi}{1-\phi}$. This establishes the second claim.

To establish the last claim, the equilibrium condition gives

$$\frac{\mathrm{EXP}_t}{\mathrm{GDP}_t} = \frac{p_{t-1}^k}{p_t^k}\frac{\mu n_{t-1}^{\alpha-1} + (\mu-1)\, n_{t-1}^{*\alpha-1}}{n_t^{\alpha-1}}$$

In steady state:

$$\begin{aligned}\frac{\mathrm{EXP}_t}{\mathrm{GDP}_t} &= \frac{1}{\mu}\frac{\mu n^{\alpha-1} + (\mu-1)\, n^{*\alpha-1}}{n^{\alpha-1}} \\ &= 1 + \frac{\mu-1}{\mu}\frac{n^{*\alpha}}{n^{\alpha}}\frac{n}{n^*} \\ &\geq 1\end{aligned}$$

with equality if and only if $\mu = 1$. A similar argument establishes $\mathrm{EXP}^*/\mathrm{GDP}^* \leq 1$.

A.3 Proof of Lemma 2

Let $K_{jt} = \sum_{i=1,2,3} k_{jt}^i$ for $j = H, F$ and $K_{Ft}^* = \sum_{i=1,2,3} k_{Ft}^{i*}$ be aggregate consumption of goods in the domestic and foreign economies, respectively. The allocation in the foreign economy must

satisfy the following conditions.

$$p_t^{c*}\left(1-n_t^*\right)^\alpha = (1-\phi)\left(p_{t-1}^{k*}n_{t-1}^{*\alpha} + p_{t-1}^{c*}\left(1-n_{t-1}^*\right)^\alpha\right)$$
$$p_t^{k*}K_{Ft}^* = \phi\left(p_{t-1}^{k*}n_{t-1}^{*\alpha} + p_{t-1}^{c*}\left(1-n_{t-1}^*\right)^\alpha\right)$$

From the optimal sectoral choice:

$$\frac{p_t^{c*}}{p_t^{k*}} = \frac{n_t^{*\alpha}}{\left(1-n_t^*\right)^\alpha}\frac{\left(1-n_t^*\right)}{n_t^*}$$

Using this:

$$n_t^{*\alpha-1} - n_t^{*\alpha} = (1-\phi)\frac{p_{t-1}^{k*}}{p_t^{k*}}n_{t-1}^{*\alpha-1}$$
$$K_{Ft}^* = \phi\frac{p_{t-1}^{k*}}{p_t^{k*}}n_{t-1}^{*\alpha-1}$$

The first equation is (15), and the second is optimal demand for foreign-produced goods in the foreign economy. Allocation in the home economy must satisfy

$$p_t^c\left(1-n_t\right)^\alpha = (1-\phi)\left(p_{t-1}^k n_{t-1}^\alpha + p_{t-1}^c\left(1-n_{t-1}\right)^\alpha + (\mu-1)M_{t-1} + \theta p_t^{k*}K_{Ft}\right)$$
$$p_t^k K_{Ht} + (1+\theta)p_t^{k*}K_{Ft} = \phi\left(p_{t-1}^k n_{t-1}^\alpha + p_{t-1}^c\left(1-n_{t-1}\right)^\alpha + (\mu-1)M_{t-1} + \theta p_t^{k*}K_{Ft}\right)$$

Replacing the equilibrium value of M_{t-1} and K_{Ht} and using the optimal sectoral choice condition, these simplify to

$$n_t^{\alpha-1} - n_t^\alpha = (1-\phi)\left(\mu\frac{p_{t-1}^k}{p_t^k}n_{t-1}^{\alpha-1} + (\mu-1)\frac{p_{t-1}^{k*}}{p_t^k}n_{t-1}^{*\alpha-1} + \theta\frac{p_t^{k*}}{p_t^k}K_{Ft}\right)$$
$$n_t^\alpha + (1+\theta)\frac{p_t^{k*}}{p_t^k}K_{Ft} = \phi\left(\mu\frac{p_{t-1}^k}{p_t^k}n_{t-1}^{\alpha-1} + (\mu-1)\frac{p_{t-1}^{k*}}{p_t^k}n_{t-1}^{*\alpha-1} + \theta\frac{p_t^{k*}}{p_t^k}K_{Ft}\right)$$

The equilibrium imports in the home country are

$$K_{Ft} = n_t^{*\alpha} - K_{Ft}^* = n_t^{*\alpha} - \phi\frac{p_{t-1}^{k*}}{p_t^{k*}}n_{t-1}^{*\alpha-1}$$

Substituting for K_{Ft} and $p_t^k = (1+\theta)p_t^{k*}$ gives equations (16) and (17).

A.4 Proof of Proposition 2

Part (i). Impose the steady-state condition $n_t^* = n^*$ in (15):

$$\frac{1-n^*}{n^*} n^{*\alpha} = (1-\phi)\frac{1}{\mu}\frac{n^{*\alpha}}{n^*},$$

which simplifies to $1 - n^* = (1-\phi)/\mu$, so $n^* = 1 - (1-\phi)/\mu$. This depends only on μ and ϕ; it is independent of θ.

Part (ii). Imposing steady state in (16) and substituting $p_t^k = (1+\theta)p_t^{k*}$, after substituting the expression for n^* and rearranging, the equilibrium condition for n reduces to

$$\phi\frac{n^\alpha}{n} - n^\alpha = (1-\phi)\left(\frac{\mu-1}{\mu}\right)\frac{n^{*\alpha}}{n^*}.$$

The right-hand side depends only on n^* (independent of θ by part (i)), μ, and ϕ. Hence n is independent of θ.

Part (iii). In steady state, foreign old-age goods demand is $k_F^* = (\phi/\mu)n^{*\alpha-1}$. Hence home imports are

$$K_F = n^{*\alpha} - \frac{\phi}{\mu}n^{*\alpha-1} = \frac{\mu-1}{\mu}\, n^{*\alpha-1},$$

which depends only on μ, ϕ, α, and n^*, all independent of θ. □

A.5 Proof of Proposition 5

Writing demand equation (27) in logs:

$$\log(k_H) = \log\left(\frac{1-\eta}{\eta}\right) + \log(k_F) + \rho\log\left(\frac{1+\theta}{q(\theta)}\right).$$

Now we consider two cases.

Case 1: Suppose $\frac{d}{d\theta}\log\left(\frac{1+\theta}{q(\theta)}\right) < 0$. Since $\partial k_F/\partial\theta < 0$ (by Lemmas 3 and 4), this implies $\partial k_H/\partial\theta < 0$. Recall from Lemma 3 that $\partial k_H^*/\partial q < 0$, and from Lemma 4 that $\partial q/\partial\theta > 0$, so $\partial k_H^*/\partial\theta < 0$. Therefore, both home and foreign demand for home-produced goods fall as the tariff rate increases, so production of home goods must fall.

Case 2: Suppose $\frac{d}{d\theta}\log\left(\frac{1+\theta}{q(\theta)}\right) \geq 0$. Therefore $\frac{1+\theta}{q(\theta)} \geq \frac{1}{q(0)}$ for any $\theta > 0$. Now consider the demand for services in the home economy, equation (26):

$$\phi n(\theta)^{\alpha-1} - n(\theta)^\alpha = (1-\phi)\left(\left(\frac{\mu-1}{\mu}\right)\frac{n^{*\alpha-1}}{q(\theta)} + \frac{\theta k_F}{q(\theta)}\right).$$

From the balance-of-payments identity (25):

$$\frac{k_F}{q(\theta)} = \left(\frac{\mu-1}{\mu}\right)\frac{n^{*\alpha-1}}{q(\theta)} + k_H^*.$$

Replacing this back into equation (26):

$$\begin{aligned}
\phi n(\theta)^{\alpha-1} - n(\theta)^{\alpha} &= (1-\phi)\left(\left(\frac{\mu-1}{\mu}\right)\frac{n^{*\alpha-1}}{q(\theta)} + \theta\left(\frac{\mu-1}{\mu}\right)\frac{n^{*\alpha-1}}{q(\theta)} + \theta k_H^*\right) \\
&= (1-\phi)\left(\left(\frac{\mu-1}{\mu}\right)\frac{1+\theta}{q(\theta)}n^{*\alpha-1} + \theta k_H^*\right) \\
&\geq (1-\phi)\left(\frac{\mu-1}{\mu}\right)\frac{1}{q(0)}n^{*\alpha-1} \\
&= \phi n(0)^{\alpha-1} - n(0)^{\alpha}.
\end{aligned}$$

Therefore, since $\phi n^{\alpha-1} - n^{\alpha}$ is decreasing in n, we have $n(\theta) \leq n(0)$. □

A.6 Proof of Proposition 3

Part (i). The foreign problem is unchanged by κ. Hence $n^* = 1 - (1-\phi)/\mu$ and $K_F = [(\mu - 1)/\mu]n^{*\alpha-1}$, both independent of κ.

Part (ii). In steady state, the home equilibrium condition is

$$H(n,\kappa) \equiv (1+\kappa)\phi n^{\alpha-1} - (1+\phi\kappa)n^{\alpha} - (1-\phi)\frac{\mu-1}{\mu}n^{*\alpha-1} = 0.$$

By the implicit function theorem, $dn/d\kappa = -H_\kappa/H_n$. We have

$$H_\kappa = \phi n^{\alpha-1} - \phi n^{\alpha} = \phi n^{\alpha-1}(1-n) > 0,$$

and

$$H_n = (1+\kappa)\phi(\alpha-1)n^{\alpha-2} - (1+\phi\kappa)\alpha n^{\alpha-1} = n^{\alpha-2}\big[(1+\kappa)\phi(\alpha-1) - (1+\phi\kappa)\alpha n\big].$$

For $n \in (0,1)$ and $\alpha \in (0,1)$ both terms in brackets are negative, so $H_n < 0$. Therefore $dn/d\kappa = -H_\kappa/H_n > 0$. □

A.7 Proof of Proposition 7

Imposing steady state in the fiscal-spending system (Appendix B, equations (34)–(35)), the labor allocations satisfy $n^* = 1 - (1-\phi)/\mu$ and $n = 1 - (1-\phi)/[\mu(1-g)]$, exactly as in the

benchmark but with endogenous μ. The money-growth rate is determined by the government budget constraint (36):

$$\big(\mu(1-g)-1\big)n^{\alpha-1}+(\mu-1)n^{*\alpha-1}=0.$$

Substituting the expressions for n and n^*, define

$$\tilde{F}(\mu)\equiv\big((1-g)\mu-1\big)\left(\mu-\frac{1-\phi}{1-g}\right)^{\alpha-1}+(\mu-1+\phi)^{\alpha-1}(\mu-1).$$

The function $\tilde{F}$ is strictly increasing on $\big(\frac{1-\phi}{1-g},\frac{1}{1-g}\big)$, with $\tilde{F}\to-\infty$ at the left endpoint and $\tilde{F}>0$ at $\mu=\frac{1}{1-g}$. By the intermediate value theorem there is a unique $\hat{\mu}\in\big(\frac{1-\phi}{1-g},\frac{1}{1-g}\big)$. One further verifies $\hat{\mu}>1$. Since θ does not appear in $\tilde{F}$, $\hat{\mu}$, n, and n^* are all independent of θ. $\square$

B Alternative Fiscal Policy

B.1 Closed Economy

Consider the same economy as in the main text. The domestic government has exogenously given expenditures and consumes a constant fraction g of the output of goods and services. This expenditure is financed by issuing nominal asset M_t. Therefore, the domestic government's budget constraint is

$$M_t - M_{t-1} = g\left(p_t^c\left(1-n_t\right)^\alpha + p_t^k n_t^\alpha\right) \tag{32}$$

Here, g is the exogenous policy choice, and the path of M_t that finances the expenditure is an endogenous equilibrium object.

Definition. *A closed economy competitive equilibrium is a sequence of worker consumption allocations and money holdings* $\{c_t^i, k_t^i, m_t^i\}_{t=0,i\in\{1,2,3\}}^\infty$*, the fraction of type 3 households in the goods sector* $\{n_t\}_{t=0}^\infty$*, prices* $\left\{p_t^c, p_t^k\right\}_{t=0}^\infty$*, wages* $\{w_t^i\}_{t=0,i\in\{1,2,3\}}^\infty$*, and government policy* $\{M_t, g\}_{t=0}^\infty$*, such that*

1. Given prices, wages and government policy, allocation $\{c_t^i, k_t^i\}_{t=0}^\infty$ solves the problem of individual worker i'

$$\max u\left(c_{t+1}^i, k_{t+1}^i\right)$$

s.t.

$$\begin{aligned} m_t^i &\leq w_t^i \\ p_{t+1}^c c_{t+1}^i + p_{t+1}^k k_{t+1}^i &\leq m_t^i \end{aligned}$$

2. The wages are determined according to equations (1), (2) and (3).

3. The government budget constraint (32) holds.

4. Markets clear

$$\begin{aligned} \sum_{i=1,2,3} c_t^i &= (1-g)\left(1-n_t\right)^\alpha \\ \sum_{i=1,2,3} k_t^i &= (1-g)\, n_t^\alpha \\ \sum_{i=1,2,3} m_t^i &= M_t \end{aligned}$$

Optimal demand by old households is given by

$$p_{t+1}^{k}\left(1-g\right)n_{t+1}^{\alpha}=\phi M_{t}$$
$$p_{t+1}^{c}\left(1-g\right)\left(1-n_{t+1}\right)^{\alpha}=\left(1-\phi\right)M_{t}$$

This implies

$$\frac{p_{t+1}^{k}}{p_{t+1}^{c}}\frac{n_{t+1}^{\alpha}}{\left(1-n_{t+1}\right)^{\alpha}}=\frac{\phi}{1-\phi}.$$

Optimal sector choice by young mobile households implies

$$\frac{p_{t+1}^{k}}{p_{t+1}^{c}}=\frac{\left(1-n_{t+1}\right)^{\alpha-1}}{n_{t+1}^{\alpha-1}}.$$

Combining these equations yields

$$n_{t+1}=\phi.$$

Total income of young households in period t equals total demand for assets:

$$M_{t}=p_{t}^{k}n_{t}^{\alpha}+p_{t}^{c}\left(1-n_{t}\right)^{\alpha}=p_{t}^{k}n_{t}^{\alpha-1},$$

where the second equality follows from optimal sector choice. Total expenditures in period $t+1$ equal the asset position of old households. Aggregate private consumption of goods and services in period $t+1$ is $\left(1-g\right)n_{t}^{\alpha}$ and $\left(1-g\right)\left(1-n_{t}\right)^{\alpha}$, respectively. Therefore,

$$M_{t}=p_{t+1}^{k}\left(1-g\right)n_{t+1}^{\alpha}+p_{t+1}^{c}\left(1-g\right)\left(1-n_{t+1}\right)^{\alpha}=p_{t+1}^{k}n_{t+1}^{\alpha-1}$$

Combining these equations and replacing $n_{t}=\phi$ gives

$$\frac{p_{t+1}^{k}}{p_{t}^{k}}=\frac{1}{1-g}.$$

and therefore

$$\mu\equiv\frac{M_{t+1}}{M_{t}}=\frac{1}{1-g}.$$

In the closed economy, money grows at a constant rate $\mu=\frac{1}{1-g}$, resulting in a constant inflation rate $\frac{p_{t+1}^{c}}{p_{t}^{c}}=\frac{p_{t+1}^{k}}{p_{t}^{k}}=\frac{1}{1-g}$.

B.2 Trade: Tariff is used to finance deficit

We now introduce a foreign country. As in the main text, only the domestic economy can issue money. The foreign economy must export goods to the domestic economy to acquire money; otherwise no production takes place in the foreign economy.

The home government levies a tariff at rate θ on imported goods. Revenue from tariffs is used to finance expenditures.

Definition. *A tariff-distorted competitive equilibrium is a sequence of domestic and foreign allocations and money holdings* $\{(c_t^i, k_{Ht}^i, k_{Ft}^i, m_t^i), (c_t^{i*}, k_{Ft}^{i*}, m_t^{i*})\}_{t=0,\, i\in\{1,2,3\}}^{\infty}$*, mobile-labor allocations* $\{n_t, n_t^*\}_{t=0}^{\infty}$*, prices* $\{p_t^c, p_t^{c*}, p_t^k, p_t^{k*}\}_{t=0}^{\infty}$*, wages* $\{w_t^i, w_t^{i*}\}_{t=0,\, i\in\{1,2,3\}}^{\infty}$*, and government policy* $\{\theta, M_t, g\}_{t=0}^{\infty}$*, such that*

1. *Given prices, wages and government policy, allocation* $\{c_t^i, k_t^i, m_t^i\}_{t=0}^{\infty}$ *solves the problem of domestic individual worker* i*'*

$$\max u\left(c_{t+1}^i, k_{Ht+1}^i + k_{Ft+1}^i\right)$$

s.t.

$$m_t^i \leq w_t^i$$
$$p_{t+1}^c c_{t+1}^i + p_{t+1}^k k_{Ht+1}^i + (1+\theta)\, p_{t+1}^{k*} k_{Ft+1}^i \leq m_t^i$$

2. *Given prices, wages, and government policy, allocation* $\{c_t^{i*}, k_t^{i*}, m_t^{i*}\}_{t=0}^{\infty}$ *solves the problem of the foreign individual worker* i*' (note: the difference is that they do not receive the transfer)*

$$\max u\left(c_{t+1}^{i*}, k_{Ft+1}^{i*}\right)$$

s.t.

$$m_t^{i*} \leq w_t^{i*}$$
$$p_{t+1}^{c*} c_{t+1}^{i*} + p_{t+1}^{k*} k_{Ft+1}^{i*} \leq m_t^{i*}$$

3. *Domestic wages are determined according to equations (1), (2) and (3) (and similarly for foreign wages).*

4. *The government budget constraint holds*

$$M_t - M_{t-1} + \theta p_t^{k*} \left(\sum_{i=1,2,3} k_{Ft}^i \right) = g \left(p_t^c \left(1 - n_t\right)^\alpha + p_t^k n_t^\alpha \right) \tag{33}$$

5. *Allocation is feasible*

$$\begin{aligned}
\sum_{i=1,2,3} c_t^i &= (1-g)\left(1-n_t\right)^\alpha \\
\sum_{i=1,2,3} c_t^{i*} &= \left(1-n_t^*\right)^\alpha \\
\sum_{i=1,2,3} k_{Ht}^i &= (1-g)\, n_t^\alpha \\
\sum_{i=1,2,3} k_{Ft}^i + \sum_{i=1,2,3} k_{Ft}^{i*} &= n_t^{*\alpha} \\
\sum_{i=1,2,3} m_t^i + \sum_{i=1,2,3} m_t^{i*} &= M_t
\end{aligned}$$

The law of one price must hold in equilibrium: $p_t^k = (1+\theta)\, p_t^{k*}$. Imposing market clearing and households' optimal choices of service consumption:

$$p_{t+1}^{c*} \left(1 - n_{t+1}^*\right)^\alpha = (1-\phi) \left(p_t^{k*} \left(n_t^*\right)^\alpha + p_t^{c*} \left(1 - n_t^*\right)^\alpha \right) \tag{34}$$

$$(1-g)\, p_{t+1}^c \left(1 - n_{t+1}\right)^\alpha = (1-\phi) \left(p_t^k n_t^\alpha + p_t^c \left(1 - n_t\right)^\alpha \right) \tag{35}$$

Imposing steady state and using optimal sector choice equations (22) and (21):

$$\begin{aligned}
\mu \left(1 - n^*\right)^\alpha &= (1-\phi) \left(1 - n^*\right)^{\alpha - 1} \\
(1-g)\, \mu \left(1 - n\right)^\alpha &= (1-\phi) \left(1 - n\right)^{\alpha - 1}
\end{aligned}$$

Therefore,

$$\begin{aligned}
n^* &= 1 - \frac{1-\phi}{\mu} \\
n &= 1 - \frac{1-\phi}{\mu\,(1-g)}
\end{aligned}$$

The labor allocation in the goods sector is always higher in the foreign economy.

To solve for the unknown growth rate μ, the aggregate demand for goods in the foreign

economy is

$$p_{t+1}^{k*} k_{Ft+1}^{*} = \phi \left(p_t^{k*} \left(n_t^*\right)^{\alpha} + p_t^{c*} \left(1 - n_t^*\right)^{\alpha} \right)$$

which simplifies to

$$k_{Ft+1}^{*} = \frac{\phi}{\mu} \left(n_t^*\right)^{\alpha-1}.$$

Home imports in steady state are

$$\left(n^*\right)^{\alpha} - k_F^* = \left(\frac{\mu - 1}{\mu}\right) \left(n^*\right)^{\alpha-1}$$

The total demand for assets by home and foreign young households must equal the supply:

$$\begin{aligned} M_t &= p_t^k n_t^{\alpha} + p_t^c \left(1 - n_t\right)^{\alpha} + p_t^{k*} \left(n_t^*\right)^{\alpha} + p_t^{c*} \left(1 - n_t^*\right)^{\alpha} \\ &= p_t^k \left(n_t^{\alpha-1} + \frac{\left(n_t^*\right)^{\alpha-1}}{1+\theta} \right) \end{aligned}$$

Substituting into the government budget constraint:

$$\begin{aligned} p_t^k \left(n_t^{\alpha-1} + \frac{\left(n_t^*\right)^{\alpha-1}}{1+\theta} \right) - p_{t-1}^k \left(n_{t-1}^{\alpha-1} + \frac{\left(n_{t-1}^*\right)^{\alpha-1}}{1+\theta} \right) \\ = p_t^k \left(g n_t^{\alpha-1} - \frac{\theta}{1+\theta} \left(\left(n_t^*\right)^{\alpha} - \frac{\phi}{\mu} \left(n_{t-1}^*\right)^{\alpha-1} \right) \right) \end{aligned}$$

Imposing steady state and rearranging:

$$\left(1 - g - \frac{1}{\mu}\right) n^{\alpha-1} + \left(\frac{\mu - 1}{\mu}\right) \left(n^*\right)^{\alpha-1} = 0 \tag{36}$$

where $n^* = 1 - \frac{1-\phi}{\mu}$ and $n = 1 - \frac{1-\phi}{\mu(1-g)}$.

The tariff has no long-term impact on allocation.

Proposition 8. *In equilibrium, there is a unique inflation rate μ such that $1 < \mu < \frac{1}{1-g}$.*

Proof. Define

$$F(\mu) = \left(\left(1 - g\right)\mu - 1\right) \left(\mu - \frac{1-\phi}{1-g}\right)^{\alpha-1} + \left(\mu - 1 + \phi\right)^{\alpha-1} \left(\mu - 1\right)$$

Step 1. The equation above has a unique solution in the interval $\left(\frac{1-\phi}{1-g}, \frac{1}{1-g}\right)$.

$F(\cdot)$ is monotonically increasing on the interval $\left(\frac{1-\phi}{1-g}, \frac{1}{1-g}\right)$:

$$\begin{aligned}
F'(\mu) &= (1-g)\left(\mu - \frac{1-\phi}{1-g}\right)^{\alpha-1} + (\alpha-1)\left((1-g)\mu - 1\right)\left(\mu - \frac{1-\phi}{1-g}\right)^{\alpha-2} \\
&+ (\mu - 1 + \phi)^{\alpha-1} + (\alpha-1)(\mu - 1 + \phi)^{\alpha-2}(\mu - 1) \\
&= \left(\mu - \frac{1-\phi}{1-g}\right)^{\alpha-2}\left((1-g)\left(\mu - \frac{1-\phi}{1-g}\right) + (\alpha-1)\left((1-g)\mu - 1\right)\right) \\
&+ (\mu - 1 + \phi)^{\alpha-2}\left((\mu - 1 + \phi) + (\alpha-1)(\mu - 1)\right) \\
&= \left(\mu - \frac{1-\phi}{1-g}\right)^{\alpha-2}\left((1-g)\mu - (1-\phi) + (\alpha-1)\left((1-g)\mu - 1\right)\right) \\
&+ (\mu - 1 + \phi)^{\alpha-2}\left(\mu - (1-\phi) + (\alpha-1)(\mu - 1)\right) \\
&> 0
\end{aligned}$$

where the last inequality holds because both terms are positive for $\mu \in \left(\frac{1-\phi}{1-g}, \frac{1}{1-g}\right)$.

Evaluating F on the boundary:

$$F\left(\frac{1-\phi}{1-g}\right) = -\phi\left(\frac{1-\phi}{1-g} - \frac{1-\phi}{1-g}\right)^{\alpha-1} + \left(\frac{1-\phi}{1-g} - 1 + \phi\right)^{\alpha-1}\left(\frac{1-\phi}{1-g} - 1\right) = -\infty$$

and

$$F\left(\frac{1}{1-g}\right) = \left(\frac{1}{1-g} - 1 + \phi\right)^{\alpha-1}\left(\frac{1}{1-g} - 1\right) > 0$$

By the intermediate value theorem, there is a unique $\hat{\mu}$ such that $F(\hat{\mu}) = 0$.

Step 2. $\hat{\mu} > 1$.

In the equation

$$\left((1-g)\mu - 1\right)\left(\mu - \frac{1-\phi}{1-g}\right)^{\alpha-1} + (\mu - 1 + \phi)^{\alpha-1}(\mu - 1) = 0$$

we know $\hat{\mu} < \frac{1}{1-g}$, so $\left((1-g)\hat{\mu} - 1\right)\left(\hat{\mu} - \frac{1-\phi}{1-g}\right)^{\alpha-1} < 0$. If $\hat{\mu} \leq 1$, then $F(\hat{\mu}) < 0$, contradicting the definition of $\hat{\mu}$. Therefore,

$$1 < \hat{\mu} < \frac{1}{1-g}.$$

□

C CES Preferences over Goods and Services

The benchmark model assumes Cobb–Douglas preferences between tradable goods and services, so that the expenditure share on tradables is a constant ϕ. This appendix shows that tariff neutrality does not depend on this unit-elasticity assumption. We replace the Cobb–Douglas aggregator with a general CES aggregator (elasticity $\varepsilon > 0$) while maintaining perfect substitutability between home and foreign tradables.

Preferences. The home economy's preferences are

$$u(k_F + k_H,\, c) = \left[\phi^{\frac{1}{\varepsilon}}(k_F + k_H)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\phi)^{\frac{1}{\varepsilon}}\, c^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}},$$

and the foreign economy's preferences are

$$u(k_F^*,\, c^*) = \left[\phi^{*\frac{1}{\varepsilon}}\,(k_F^*)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\phi^*)^{\frac{1}{\varepsilon}}\, c^{*\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}},$$

where ε is the elasticity of substitution between goods and services, ϕ and ϕ^* are the goods-weight parameters, and the remaining notation is as in the main text. The benchmark Cobb–Douglas case corresponds to $\varepsilon = 1$.

Equilibrium. The equilibrium definition is analogous to the benchmark. Producer optimality and optimal sectoral choice by mobile workers continue to imply $p^k/p^c = [(1-n)/n]^{\alpha-1}$ and $p^{k*}/p^{c*} = [(1-n^*)/n^*]^{\alpha-1}$. With CES preferences, the demand by old workers for services in the foreign economy is

$$p_{t+1}^{c*} c_{t+1}^* = \frac{1-\phi^*}{1-\phi^* + \phi^*\left(\frac{p_{t+1}^{k*}}{p_{t+1}^{c*}}\right)^{1-\varepsilon}}\, p_t^{k*} n_t^{*\alpha-1},$$

and the demand for goods is

$$p_{t+1}^{k*} k_{Ft+1}^* = \frac{\phi^*\left(\frac{p_{t+1}^{k*}}{p_{t+1}^{c*}}\right)^{1-\varepsilon}}{1-\phi^* + \phi^*\left(\frac{p_{t+1}^{k*}}{p_{t+1}^{c*}}\right)^{1-\varepsilon}}\, p_t^{k*} n_t^{*\alpha-1}.$$

Imposing steady state, using market clearing for services ($c^* = (1-n^*)^\alpha$), and substituting the optimal sectoral choice condition for p^{k*}/p^{c*}, the foreign labor allocation n^* solves

$$1 - n^* = \frac{1-\phi^*}{1-\phi^* + \phi^* \left(\left(\frac{1-n^*}{n^*} \right)^{\alpha-1} \right)^{1-\varepsilon}} \frac{1}{\mu}. \tag{37}$$

This equation does not admit a closed-form solution in general, but the key observation is that it does not involve θ: the foreign labor allocation is independent of the tariff.

Import quantity. The balance-of-payments identity is unchanged: $p^{k*}_{t+1} k_{Ft+1} = (\mu - 1) p^{k*}_t n^{*\alpha-1}_t$. In steady state,

$$k_F = \frac{\mu - 1}{\mu} n^{*\alpha-1},$$

which is invariant to θ, exactly as in the benchmark.

Home labor allocation. The demand by old workers for services in the home economy is

$$p^c_{t+1} c_{t+1} = \frac{1-\phi}{1-\phi + \phi \left(\frac{p^k_{t+1}}{p^c_{t+1}} \right)^{1-\varepsilon}} \left(\mu p^k_t n^{\alpha-1}_t + (\mu-1) p^{k*}_t n^{*\alpha-1}_t + \theta\, p^{k*}_{t+1} k_{Ft+1} \right).$$

In steady state, imposing market clearing for services ($c = (1-n)^\alpha$), substituting the optimal sectoral choice condition for p^k/p^c, and noting that with perfect substitutability the law of one price gives $p^k = (1+\theta) p^{k*}$ (so that $q \equiv p^k/p^{k*} = 1+\theta$), we obtain

$$\left[1 - \phi + \phi \left(\left(\frac{1-n}{n} \right)^{\alpha-1} \right)^{1-\varepsilon} \right] (n^{\alpha-1} - n^\alpha) = (1-\phi) \left(n^{\alpha-1} + \frac{1}{1+\theta} \left(\frac{\mu-1}{\mu} \right) n^{*\alpha-1} + \frac{\theta}{1+\theta} k_F \right). \tag{38}$$

Substituting $k_F = \frac{\mu-1}{\mu} n^{*\alpha-1}$ from the balance-of-payments identity into the right-hand side:

$$\begin{aligned} \frac{1}{1+\theta} \left(\frac{\mu-1}{\mu} \right) n^{*\alpha-1} + \frac{\theta}{1+\theta} \cdot \frac{\mu-1}{\mu} n^{*\alpha-1} &= \frac{1+\theta}{1+\theta} \left(\frac{\mu-1}{\mu} \right) n^{*\alpha-1} \\ &= \left(\frac{\mu-1}{\mu} \right) n^{*\alpha-1}. \end{aligned}$$

The tariff rate θ cancels exactly. Equation (38) therefore reduces to

$$\left[1 - \phi + \phi \left(\left(\frac{1-n}{n} \right)^{\alpha-1} \right)^{1-\varepsilon} \right] (n^{\alpha-1} - n^\alpha) = (1-\phi) \left(n^{\alpha-1} + \left(\frac{\mu-1}{\mu} \right) n^{*\alpha-1} \right), \tag{39}$$

which pins down n independently of θ.

Proposition 9 (Tariff Neutrality with CES Goods–Services Preferences)**.** *Suppose preferences over goods and services are CES with elasticity $\varepsilon > 0$, and home and foreign tradables are perfect substitutes. The steady-state labor allocations (n, n^*) and the import quantity k_F are invariant to the tariff rate θ. The Cobb–Douglas expenditure-share assumption is not necessary for tariff neutrality.*

Proof. Equation (37) determines n^* independently of θ. The balance-of-payments identity then pins down k_F independently of θ. Equation (39) determines n independently of θ, as shown by the cancellation above. □

The mechanism is the same as in the benchmark: tariff revenue is rebated to home consumers, and with perfect substitutability the law of one price ties the producer price ratio to $1 + \theta$. These two facts together ensure that θ drops out of the home equilibrium condition. The CES weights ϕ, ϕ^*, and the elasticity ε affect the *levels* of n and n^* but do not restore any role for the tariff.

D Multi-Country Environment with Portfolio Choice

This appendix extends the benchmark model by relaxing the assumption that the home transaction asset is the only store of value and that the foreign economy is a single country. We develop a multi-country environment with differentiated goods, a foreign settlement currency, and a micro-founded portfolio problem. Tariff neutrality survives: the import tariff θ does not appear in any steady-state allocation. What *does* affect allocations is financial policy—a tax on foreign reserve holdings—which we contrast with the goods tariff.

Notation. We maintain the notation of the main paper throughout. In particular: ϕ is the Cobb–Douglas expenditure share on goods, α is the production-function exponent, $\mu \geq 1$ is the dominant-asset growth rate, n and n^* are the mobile labor shares in the goods sector at home and abroad, and $\theta \geq 0$ is the ad-valorem import tariff. To avoid confusion between the two currencies in this extension, we refer to the home currency as the *dominant asset* throughout this appendix.

D.1 Countries and demographics

There is one *home* country and a unit mass of *foreign* countries indexed by $i \in [0, 1]$. All countries share the same OLG structure: agents live for two periods, the young work, the old consume.

Foreign countries. Each foreign country i has three types of workers: type 1 specific to goods variety i, type 2 specific to services, type 3 mobile across the two sectors. There is one unit of each type.

Home country. The home country has a richer labor structure reflecting its ability to produce all varieties. There is a *continuum* of type 1 workers indexed by $i \in [0, 1]$, one for each goods variety i. Each type-1_i worker is specific to variety i. There is one unit of type 2 workers (specific to services) and one unit of type 3 workers (mobile). The mobile workers choose which variety to produce, allocating themselves across the continuum of goods industries and the service sector.

D.2 Goods, production, and preferences

Differentiated goods. There is a unit mass of tradable good varieties indexed by $i \in [0, 1]$. Foreign country i can produce only variety i. The home country can produce *all* varieties $i \in [0, 1]$. The asymmetry between home and foreign is therefore in production capacity (home is diversified, foreign is specialized) and in the financial system (home issues the reserve asset), not in preferences.

Services are non-tradable and homogeneous within each country.

Production. In each foreign country i, variety i is produced using the type-1 worker and any mobile labor allocated to goods: output is $n^{*\alpha}$, where n^* is the mobile labor in goods. Service output is $(1-n^*)^\alpha$.

At home, each variety i is produced using type-1_i (the fixed factor for variety i) and a measure ν_i of mobile labor: output of variety i is ν_i^α. Mobile workers are freely mobile across all goods industries and services, so in equilibrium they earn the same wage everywhere. Service output is $(1-n)^\alpha$, where $n = \int_0^1 \nu_i \, di$ is total mobile labor in goods.

Preferences. All agents—home and foreign—have identical preferences. Old agents consume services and a symmetric Dixit-Stiglitz aggregate over goods varieties:

$$K = \left[\int_0^1 k_i^{(\varepsilon-1)/\varepsilon} \, di \right]^{\varepsilon/(\varepsilon-1)}, \tag{40}$$

where k_i is consumption of variety i and $\varepsilon > 1$ is the elasticity of substitution. The overall utility is Cobb–Douglas:

$$U = K^\phi \, c^{1-\phi}.$$

Trade pattern. Since each foreign country produces only one variety but consumes all, foreign countries must trade with each other and with home. The home country produces all varieties, so it *can* self-supply goods—but in equilibrium it also imports foreign varieties (since foreign varieties are perfect substitutes for domestically produced ones within each variety, the law of one price holds variety by variety). The home country's net import position reflects the exorbitant privilege mechanism, not a production disadvantage.

D.3 Currencies

Dominant asset. The home government issues the dominant asset (fiat money) $M_t = \mu M_{t-1}$, $\mu \geq 1$, with seigniorage $\tau_t = (1 - 1/\mu) M_t$ distributed lump-sum to home old agents.

Foreign settlement instrument. Foreign countries collectively maintain a settlement system. It can be interpreted as a composite of bilateral credit arrangements, a secondary vehicle currency (for example, a regional anchor currency), or a multilateral clearing mechanism. We represent it as a single "foreign money" $\hat{M}_t = \hat{\mu} \, \hat{M}_{t-1}$, with $\hat{\mu} \geq 1$ and seigniorage distributed to foreign old agents. The exchange rate is e_t (units of the dominant asset per unit of foreign money).

Home agents hold only the dominant asset (they face no transaction disadvantage domestically and receive seigniorage). Foreign agents allocate their savings between the dominant asset and foreign money.

D.4 Transaction structure and the portfolio problem

We now connect the variety structure to a multi-market transaction problem that micro-founds the portfolio choice.

Markets as varieties. A foreign old agent must purchase a continuum of goods varieties and services. Each purchase is a *transaction* that requires currency. The key friction is that different transactions are embedded in different payment networks:

- **Goods variety** $j \neq i$ (cross-border): Country i must buy variety j from country j (or from home). This is a cross-border transaction. A fraction of such transactions—commodity trades, large-volume invoicing, contracts with distant partners—are settled in the reserve currency. The remainder—regional trade, familiar bilateral partners—can be settled through the foreign system.
- **Services** (domestic): Purchased from local providers, settled entirely in local currency.

Dominant-asset acceptance across markets. Index all markets by $j \in [0, 1]$, where a fraction ϕ of the unit interval represents goods varieties and the remaining $1 - \phi$ represents service transactions. Each market j has a dominant-asset acceptance rate $\omega(j) \in [0, 1]$: a fraction $\omega(j)$ of sellers accept the dominant asset at face value, while $1 - \omega(j)$ accept only foreign money.

Since each foreign country is measure zero, its own-variety demand is negligible. Essentially *all* of a foreign agent's goods spending (fraction ϕ) involves cross-border trade. Within this cross-border trade, the distribution of $\omega(j)$ reflects the settlement structure:

- Markets for commodities, standardized intermediates, and trade with distant partners have $\omega(j)$ near 1 (reserve currency required).
- Markets for regional goods, familiar bilateral trade have $\omega(j)$ near 0 (local settlement system suffices).
- Service markets have $\omega(j) = 0$ (purely domestic).

Effective purchasing power in each market. The agent allocates dominant-asset balances a_j and foreign-money balances b_j to each market j. Her effective purchasing power in market j is:

$$\ell_j = \omega(j)\, R^H a_j + [1 - \omega(j)]\, R^F b_j\,,$$

where $R^H = 1/\mu$ and $R^F = 1/\hat{\mu}$ are the real returns on the dominant asset and foreign money, respectively.

In each market j, the agent converts raw balances into effective purchasing power $v(\ell_j)$, where v is a concave transaction technology ($v' > 0$, $v'' < 0$, $v(0) = 0$). The concavity reflects frictions in each payment channel: congestion, search costs, or conversion inefficiencies that make the marginal unit of liquidity directed at a given market less effective than the first.

Total effective purchasing power is:

$$L = \int_0^1 v(\ell_j)\, dj\,.$$

This is the agent's *budget*, not her utility. She then allocates L across the goods aggregate and services: $\max K^{\phi} c^{1-\phi}$ subject to $P^K K + p^{c*} c = L$, where P^K is the Dixit-Stiglitz price index. Note that the pooled budget L allows liquidity generated in cross-border goods markets to fund service consumption; this separability is a tractability device that abstracts from settlement constraints at the point of sale. Because preferences are linear in L, the young agent's portfolio problem reduces to maximizing L subject to $a + b = w^*$.

Optimal currency allocation. The agent allocates total dominant-asset balances $a = \int a_j\, dj$ and foreign-money balances $b = \int b_j\, dj$ across markets. By concavity of v, each market is optimally served by the currency with the higher effective return. Market j is settled in the dominant asset if $\omega(j)\, R^H > [1 - \omega(j)]\, R^F$, i.e., if

$$\omega(j) > \frac{R^F}{R^H + R^F} = \frac{\mu}{\mu + \hat{\mu}} \equiv \bar{\omega}\,.$$

From the distribution of markets to CES. Let $G(\omega)$ denote the CDF of dominant-asset acceptance rates across markets, with density $g(\omega)$. The dominant-asset portfolio share is $s = S(\bar{\omega}; G)$, a decreasing function of $\bar{\omega}$. Under the isoelastic specification $v(\ell) = \ell^{(\sigma-1)/\sigma}/[(\sigma-1)/\sigma]$ and a distribution $g(\omega)$ that places mass $\lambda > 1/2$ on high-ω markets (reflecting the reserve currency's dominance in international settlement), the aggregate liquidity function takes the CES form:

$$L(R^H a,\, R^F b) = \left[\lambda\, (R^H a)^{\frac{\sigma-1}{\sigma}} + (1-\lambda)\, (R^F b)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}. \tag{41}$$

The parameter λ inherits its meaning from the trade structure: it is the effective mass of cross-border transactions for which the reserve currency has a comparative advantage. The parameter σ governs the curvature of the transaction technology within each market—high σ means liquidity is easily substitutable at the intensive margin (diminishing returns set in slowly), low σ means each unit of the wrong currency is a poor substitute for the right one.[5]

Remark. The results that follow hold for any smooth, homogeneous-of-degree-one aggregator $L(a, b)$ with $L_a, L_b > 0$ and strict quasi-concavity, not just CES. We use CES for closed-form expressions.

D.5 Home country

The home country can produce all varieties and services. Its preferences are the same as any foreign country. The asymmetries are: (i) home issues the reserve currency, and (ii) home has a type-1 worker for every variety.

Young agents: labor allocation. Let p_i^k denote the price of variety i. Each type-1_i worker is the fixed factor in variety i and earns the residual after paying mobile labor. Mobile workers choose which industry to work in. Free mobility of type 3 across all goods industries and services requires a common mobile wage:

$$w^3 = p_i^k \, \alpha \, \nu_i^{\alpha-1} = p^c \, \alpha \, (1-n)^{\alpha-1} \quad \text{for all } i \in [0,1].$$

This pins down the allocation of mobile labor across varieties. Since $\nu_i = (p_i^k/p_j^k)^{1/(1-\alpha)}\nu_j$ for any two varieties, symmetric prices ($p_i^k = p^k$ for all i) imply $\nu_i = n$ for all i: mobile labor is spread uniformly across varieties. In this case, home output of each variety is n^α and total home goods output is $\int_0^1 n^\alpha \, di = n^\alpha$.

Wages are then:

$$\begin{aligned} w_i^1 &= p^k(1-\alpha)\, n^\alpha \quad \text{for each } i \in [0,1], \\ w^2 &= p^c(1-\alpha)\,(1-n)^\alpha, \\ w^3 &= p^k \alpha \, n^{\alpha-1} = p^c \alpha \, (1-n)^{\alpha-1}. \end{aligned} \tag{42}$$

[5] An equivalent CES aggregator can be derived from the extensive margin by assuming a linear transaction technology $v(\ell) = \ell$ and a Fréchet (Type II extreme value) distribution over market-specific currency productivity. Under that formulation, σ directly reflects the dispersion of settlement requirements across markets, with high σ corresponding to many markets that are near-indifferent between currencies. The two approaches are mathematically isomorphic, but the isoelastic $v(\ell)$ shortcut avoids the additional algebra required by the discrete-choice derivation.

The mobile-labor indifference condition (42) pins down the relative price of goods to services:

$$\frac{p^k}{p^c} = \frac{(1-n)^{\alpha-1}}{n^{\alpha-1}} = \left(\frac{1-n}{n}\right)^{\alpha-1}.$$

Total home wage income is $W = \int_0^1 w_i^1 \, di + w^2 + w^3 = p^k(1-\alpha)n^\alpha + p^c(1-\alpha)(1-n)^\alpha + w^3$. All income is saved in the dominant asset.

Old agents. Home old agents receive their money holdings plus seigniorage $\tau_t/3$, where the transfer is divided equally among the continuum of type-1 agents, plus the single type-2 and type-3 agents (we normalize so each "type" receives $\tau_t/3$ of total seigniorage, treating the continuum of type-1 workers as one group). They maximize $K^\phi c^{1-\phi}$ subject to $p^k K + p^c c = \Omega^i$, where Ω^i is old-age wealth and K is the Dixit-Stiglitz aggregate (40). By symmetry, the price index is $P^K = p^k$.

Home old agents face no currency friction: all their balances are in the dominant asset, which is universally accepted. Their effective purchasing power equals their gross wealth.

Demand for varieties. With symmetric varieties and a common price p^k, home demand for each variety is uniform: $k_i = K$ for all i. Home agents are indifferent between domestically produced and imported units of each variety.

Seigniorage. The total stock of the dominant asset is $M_t = W_t + s\,W_t^*$ (home savings plus aggregate foreign reserve holdings), so seigniorage is $\tau_t = (\mu - 1)(W_{t-1} + s\,W_{t-1}^*)$.

D.6 Representative Foreign Country

By symmetry, every foreign country i faces the same problem. We describe a representative foreign country, dropping the index i.

Young agents. Foreign country i produces variety i and services. Let p^{k*} denote the price of any variety in terms of the dominant asset (by symmetry $p^{k*} = p^k$ under the law of one price). Wages are:

$$\begin{aligned}
w^{1*} &= p^{k*}(1-\alpha)\, n^{*\alpha}, \\
w^{2*} &= p^{c*}(1-\alpha)\,(1-n^*)^\alpha, \\
w^{3*} &= p^{k*}\alpha\, n^{*\,\alpha-1} = p^{c*}\alpha\,(1-n^*)^{\alpha-1}.
\end{aligned}$$

Total foreign wage income (per country) is $W^* = w^{1*} + w^{2*} + w^{3*}$.

Portfolio problem. Each foreign young agent saves her entire wage. She allocates savings between the dominant asset (amount a) and foreign money (amount $b = w^{i*} - a$) to maximize effective purchasing power $L(R^H a,\ R^F b)$ from the transaction problem described above. In the deterministic steady state, $R^H = 1/\mu$ and $R^F = 1/\hat{\mu}$. The CES first-order condition gives the dominant-asset portfolio share:

$$s \equiv \frac{a}{a+b} = \frac{\left(\frac{\lambda}{1-\lambda}\right)^{\sigma}\left(\frac{\hat{\mu}}{\mu}\right)^{\sigma-1}}{1+\left(\frac{\lambda}{1-\lambda}\right)^{\sigma}\left(\frac{\hat{\mu}}{\mu}\right)^{\sigma-1}}. \tag{43}$$

Three properties: s is increasing in λ (stronger dominant-asset advantage), increasing in $\hat{\mu}/\mu$ when $\sigma > 1$ (higher foreign inflation makes foreign money less attractive), and—crucially—independent of the goods tariff θ. When $\lambda = 1$ or $\hat{\mu}/\mu \to \infty$, $s \to 1$ and we recover the benchmark.

Old agents. A foreign old agent has nominal wealth $W^*_{t-1} + \hat{\tau}_t$ (savings plus foreign seigniorage). She maximizes $K^{\phi}c^{1-\phi}$ subject to $p^{k*}K + p^{c*}c = W^*_{t-1} + \hat{\tau}_t$, where K is the Dixit-Stiglitz aggregate.

Demand for varieties. With symmetric varieties and a common price, foreign demand is spread uniformly: $k_j = K$ for all j. Since foreign country i is measure zero, essentially *all* of its goods spending goes to other countries' varieties—requiring cross-border settlement. This is the source of the currency friction.

Foreign seigniorage. Foreign old agents receive foreign seigniorage. Using the foreign money-market clearing condition, $\hat{\tau}_t = (1 - 1/\hat{\mu})(1 - s)W^*_t$.

D.7 Equilibrium

Definition 4 (Multi-country equilibrium)**.** *A stationary equilibrium consists of labor allocations* $(n, \{\nu_i\}_{i\in[0,1]}, n^*)$*, prices* $(p^k, p^{k*}, p^c, p^{c*})$*, a portfolio share* s*, money supplies* $(M_t, \hat{M}_t)$*, and consumption allocations such that:*

1. ***Home household optimization.*** *Each home young agent saves all income in the dominant asset. Each home old agent maximizes* $K^{\phi}c^{1-\phi}$ *subject to* $p^c c + p^k K = \Omega^i$*.*

2. ***Home mobile labor allocation.*** *Mobile workers are indifferent across all goods industries*

and services:

$$p^k \alpha\, \nu_i^{\alpha-1} = p^c \alpha\, (1-n)^{\alpha-1} \quad \textit{for all } i \in [0,1], \qquad n = \textstyle\int_0^1 \nu_i\, di.$$

By symmetry $\nu_i = n$ for all i, which gives $p^k/p^c = [(1-n)/n]^{\alpha-1}$.

3. ***Foreign household optimization.*** *Each foreign young agent saves all income, allocating fraction s to the dominant asset and $1-s$ to foreign money, where s maximizes effective purchasing power (41). Each foreign old agent maximizes $K^{\phi} c^{1-\phi}$ subject to $p^{c*}c + p^{k*}K = W^*_{t-1} + \hat{\tau}_t$.*

4. ***Foreign labor market clearing.*** *In each foreign country, the mobile wage equals the marginal product in both sectors:*

$$w^{3*} = p^{k*} \alpha\, n^{*\,\alpha-1} = p^{c*} \alpha\, (1-n^*)^{\alpha-1}.$$

5. ***Services market clearing.*** *Services are non-tradable:*

$$\textstyle\sum_i c^i = (1-n)^{\alpha} \ \textit{(home)}, \qquad \sum_i c^{i*} = (1-n^*)^{\alpha} \ \textit{(each foreign country)}.$$

6. ***Global goods market clearing.*** *For each variety j, world demand equals world supply. By symmetry, this reduces to a single aggregate condition:*

$$n^{\alpha} + n^{*\alpha} = \phi[\textstyle\sum_i \Omega^i + \sum_i \Omega^{i*}]/p^k,$$

where the left side is world goods output (home n^{α} plus aggregate foreign $n^{*\alpha}$) and the right side is world goods demand.

7. ***Dominant-asset market clearing.*** $\sum_i m^i + s\, W^* = M_t$.

8. ***Foreign money market clearing.*** $(1-s)\, W^* = e_t \hat{M}_t$.

9. ***Government budget constraints.*** $\tau_t = M_t - M_{t-1}$ *(home);* $\hat{\tau}_t = e_t(\hat{M}_t - \hat{M}_{t-1})$ *(foreign).*

D.8 Reduction to the two-country system

The multi-country structure collapses to the two-country model of the benchmark under the symmetry assumptions.

Under symmetry, every foreign country is identical: same n^*, same wages, same portfolio share s. All varieties have the same price, so the Dixit-Stiglitz price index is $P^K = p^k$ and

variety-level demands are uniform. Cobb–Douglas upper-tier preferences therefore imply the same aggregate expenditure shares as in the benchmark: fraction ϕ of old-age expenditure is devoted to tradables and fraction $1-\phi$ to services.

The only new object relative to the benchmark is the portfolio share s. In the benchmark, all foreign saving is placed in the dominant asset. Here, only the fraction s is. That means the reserve-demand mechanism is unchanged except for a scaling: every benchmark term that comes from foreign accumulation of the dominant asset is multiplied by s, while the remaining share $1-s$ is held in the foreign settlement instrument and does not generate a real transfer to the home country.

This delivers the benchmark steady-state system with the reserve-demand term scaled by s. On the foreign side, the service-market condition gives the goods-sector labor share as the autarky allocation ϕ plus the extra tradables production needed to finance dominant-asset accumulation:

$$n^* = \phi + s\,(1-\phi)\,\frac{\mu-1}{\mu}\,, \tag{44}$$

$$\tag{45}$$

Foreign excess tradables output then equals the real transfer required to support those dominant-asset holdings:

$$K_F = s\,\frac{\mu-1}{\mu}\;n^{*\,\alpha-1}\,. \tag{46}$$

Finally, the home side is the same labor-allocation condition as in the benchmark, with that scaled real transfer on the right-hand side:

$$\phi\,n^{\alpha-1} - n^{\alpha} = s\,(1-\phi)\,\frac{\mu-1}{\mu}\;n^{*\,\alpha-1}\,. \tag{47}$$

The portfolio share s is given by (43) and depends on $(\lambda, \sigma, \mu, \hat{\mu})$. This is the precise sense in which the multi-country model reduces to the two-country benchmark under symmetry: the structure of the equilibrium system is unchanged, and the only modification is that foreign reserve demand is scaled by the endogenous portfolio share s.

What the multi-country structure adds. The multi-country environment does not change the equilibrium allocations, but it provides three things the two-country model cannot:

1. A *reason* for foreign countries to trade with each other: each produces one variety but consumes all, so cross-border goods trade is essential.

2. A *micro-foundation* for reserve-currency demand: the multi-market shopping problem arises from the variety structure—each cross-border purchase is a transaction that may require the reserve currency for settlement.

3. A *structural interpretation* of λ and σ: λ is the effective mass of cross-border goods transactions settled in the reserve currency, determined by invoicing conventions and payment-network effects; σ governs the curvature of the transaction technology within each market.

D.9 Tariff policy

We now introduce a tariff in the multi-country environment and show that tariff neutrality survives.

Tariff. The home government imposes an ad-valorem import tariff $\theta \geq 0$ on all goods varieties imported from foreign countries. The tariff creates a wedge between the price paid by home consumers and the price received by foreign producers. Let p^{k*} denote the producer price of any variety (common across foreign countries by symmetry). Home consumers pay $p^k = (1+\theta)\, p^{k*}$ for imported varieties.

Since home also produces all varieties, the domestic price of all goods—whether home-produced or imported—is $p^k = (1+\theta)\, p^{k*}$ (the tariff-inclusive price, which domestic producers also receive by the law of one price within the home market).

Tariff revenue is $T = \theta\, p^{k*}\, K_F$, where K_F is total imports (aggregate foreign goods absorbed by home). Revenue is rebated lump-sum to home old agents.

Home allocation under the tariff. The home mobile-labor condition continues to equate marginal products across sectors at *domestic* prices:

$$p^k\, \alpha\, n^{\alpha-1} = p^c\, \alpha\, (1-n)^{\alpha-1},$$

which pins down $p^k/p^c = [(1-n)/n]^{\alpha-1}$ as before. The tariff does not enter this condition: home producers face the tariff-inclusive goods price p^k in both sectors.

Home old agents have wealth $\Omega^i = w^i_{t-1} + \tau_t/3 + T/3$ (including tariff revenue). They spend fraction ϕ on goods (at price p^k) and $1-\phi$ on services (at price p^c).

Foreign allocation under the tariff. Foreign producers sell at price $p^{k*} = p^k/(1+\theta)$. But foreign agents do not pay the tariff: they buy foreign varieties at producer prices p^{k*} (cross-border trade among foreign countries is not subject to the home tariff). Their wages, portfolio allocation, and consumption decisions are unchanged:

- The portfolio share s depends on $(\lambda, \sigma, \mu, \hat{\mu})$ and does not contain θ.
- Foreign service-market clearing depends on s, μ, $\hat{\mu}$, and ϕ, not on θ.
- The foreign labor allocation n^* is independent of θ.

Home service-market clearing. Let Ω denote total home old-age wealth. Because tariff revenue is rebated lump-sum,

$$\Omega = M_t - sW^*_{t-1} + T, \qquad T = \theta p^{k*} K_F,$$

where $M_t - sW^*_{t-1}$ is the home share of the dominant asset (netting out foreign holdings). Home expenditure shares imply

$$p^c(1-n)^\alpha = (1-\phi)\Omega, \tag{48}$$

$$p^k\left(n^\alpha + K_F\right) = \phi\Omega, \tag{49}$$

where $p^k = (1+\theta)p^{k*}$ is the domestic goods price and $n^\alpha + K_F$ is total home absorption of tradables. Eliminating Ω between (48) and (49) gives

$$n^\alpha + K_F = \frac{\phi}{1-\phi}\frac{p^c}{p^k}(1-n)^\alpha.$$

Using the home labor-market condition $p^k/p^c = [(1-n)/n]^{\alpha-1}$, equivalently $p^c/p^k = (n/(1-n))^{\alpha-1}$, this reduces to

$$\phi n^{\alpha-1} - n^\alpha = (1-\phi)K_F. \tag{50}$$

Hence the home labor allocation depends on the tariff only through the import quantity K_F.

Proposition 10 (Tariff neutrality in the multi-country environment)**.** *For any ad-valorem import tariff $\theta \geq 0$ with revenue rebated lump-sum to home old agents:*

(i) The portfolio share s is independent of θ.

(ii) The foreign labor allocation n^* is independent of θ.

(iii) The home labor allocation n is independent of θ.

(iv) The aggregate import quantity K_F is independent of θ.

Proof. The argument has three steps.

Step 1: s is independent of θ. The portfolio share is determined by the CES first-order condition (41), which depends on asset returns $(R^H, R^F) = (1/\mu, 1/\hat{\mu})$ and the transaction-structure parameters (λ, σ). The tariff θ creates a wedge in goods prices but does not alter any asset return. Therefore s does not depend on θ.

Step 2: n^* and K_F are independent of θ. By the reduction to the two-country system in (44)–(46),

$$n^* = \phi + s(1-\phi)\frac{\mu-1}{\mu}, \qquad K_F = s\frac{\mu-1}{\mu}n^{*\,\alpha-1}.$$

Both objects depend on the tariff only through s, and Step 1 shows that s is independent of θ. Hence both n^* and K_F are independent of the tariff.

Step 3: n is independent of θ. The home-side equilibrium condition is (50),

$$\phi n^{\alpha-1} - n^{\alpha} = (1-\phi)K_F.$$

The left-hand side depends only on n, while the right-hand side depends on the tariff only through K_F. By Step 2, K_F is independent of θ. Therefore the equation determining n is independent of the tariff, so the home labor allocation is independent of θ as well.

The intuition is the same as in the benchmark. The tariff is a goods-price wedge. It cannot alter the foreign portfolio share s, the foreign labor allocation n^*, or the net real transfer K_F, because those are pinned down by the foreign demand for the reserve asset. Once K_F is fixed, the home goods-market and service-market conditions imply the same labor allocation as in the no-tariff economy. □

Remark (Tariff and the Dixit-Stiglitz structure). The tariff applies to home imports of all foreign varieties. With symmetric varieties and a common price, it does not distort the composition of imports—home still demands each foreign variety equally. The tariff contracts the total volume of imports *only if* it changes the net real transfer K_F. Since K_F is pinned by the reserve-demand channel, the tariff does not contract import volume. The measured trade deficit $p^{k*}K_F/(p^k n^\alpha + p^c(1-n)^\alpha)$ changes mechanically through the price wedge (a valuation effect), but the real quantity K_F does not.

D.10 Financial tariff

While the goods tariff is neutral, a tax on foreign reserve holdings operates on the asset-return margin and therefore *does* change allocations.

Setup. Suppose the home government imposes a tax $\tau_f \in [0, 1)$ on foreign holdings of the dominant asset. The effective real return on the dominant asset for foreign agents becomes $R^H = (1 - \tau_f)/\mu$ instead of $1/\mu$. The return on foreign money $R^F = 1/\hat{\mu}$ is unchanged.

Modified portfolio share. Substituting the after-tax return into the CES first-order condition, the portfolio share becomes:

$$s(\tau_f) = \frac{\left(\frac{\lambda}{1-\lambda}\right)^{\sigma} \left(\frac{\hat{\mu}(1-\tau_f)}{\mu}\right)^{\sigma-1}}{1 + \left(\frac{\lambda}{1-\lambda}\right)^{\sigma} \left(\frac{\hat{\mu}(1-\tau_f)}{\mu}\right)^{\sigma-1}} . \tag{51}$$

For $\sigma > 1$, s is strictly decreasing in τ_f: the financial tax makes the dominant asset less attractive, shifting foreign portfolios toward foreign money.

Effect on allocations. Since the steady-state conditions are:

$$n^* = \phi + s\,(1 - \phi)\,\frac{\mu - 1}{\mu}\,, \tag{52}$$

$$\phi\, n^{\alpha-1} - n^{\alpha} = s\,(1 - \phi)\,\frac{\mu - 1}{\mu}\; n^{*\,\alpha-1}\,, \tag{53}$$

$$K_F = s\,\frac{\mu - 1}{\mu}\; n^{*\,\alpha-1}\,, \tag{54}$$

a reduction in s propagates through all three conditions.

Proposition 11 (Financial tariff effectiveness)**.** *For $\sigma > 1$, an increase in the reserve-accumulation tax τ_f:*

(i) Reduces the portfolio share: $\partial s/\partial \tau_f < 0$.

(ii) Reduces foreign goods-sector labor: $\partial n^*/\partial \tau_f < 0$.

(iii) Increases home goods-sector labor: $\partial n/\partial \tau_f > 0$.

(iv) Reduces home imports: $\partial K_F/\partial \tau_f < 0$.

Proof. Part (i): In (51), $(1 - \tau_f)$ enters with exponent $\sigma - 1 > 0$ in the numerator. Increasing τ_f reduces the numerator relative to the denominator.

Part (ii): From (52), n^* is increasing in s. Lower s gives lower n^*.

Part (iv): From (54), K_F depends on s directly and through n^*. The direct effect reduces K_F; the indirect channel (lower n^* raises $n^{*\,\alpha-1}$ since $\alpha < 1$) works in the opposite direction but is dominated, so K_F falls.

Part (iii): The right-hand side of (53) falls when s falls: the direct effect of lower s dominates the opposing indirect channel through $n^{*\alpha-1}$. The left-hand side $\phi n^{\alpha-1} - n^{\alpha}$ is decreasing in n for $n < \phi$. A smaller right-hand side therefore requires a larger n. □

Interpretation. The financial tariff succeeds where the goods tariff fails. It reduces the trade deficit and reverses deindustrialization. The mechanism is direct: by taxing the return on the reserve asset, the home government makes foreign money relatively more attractive, reducing foreign demand for the dominant asset (s falls). With less foreign savings flowing into the home asset, fewer goods flow from foreign to home in return, and the net real transfer K_F shrinks. On the production side, the home tradables sector expands (n rises) because the exorbitant-privilege drag is reduced.

Contrast with goods tariff. The goods tariff θ and the financial tariff τ_f target different margins:

Policy instrument	Affects allocations?	Operates on
Import tariff θ	No	Goods-price margin
Financial tariff τ_f	Yes	Asset-return margin
Monetary policy μ	Yes	Asset-return margin

The trade deficit in a reserve-currency economy is pinned by foreign demand for the home asset. Policies that change the attractiveness of the home asset (financial taxes, monetary policy) can alter the trade deficit; policies that change goods prices (tariffs) cannot. This orthogonality is not an artifact of the benchmark's single-currency assumption—it survives in the multi-country environment with differentiated goods, a foreign settlement currency, and endogenous portfolio choice.

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